Anas_Justin_Nate_CSE4502_Project

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1 Importing Libraries

[470]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from sklearn import preprocessing
from pandas.plotting import scatter_matrix
import graphviz
from sklearn import tree
import pydotplus
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn import tree
from IPython.display import Image
import scikitplot as skplt
import sklearn as sk
from sklearn.preprocessing import StandardScaler
import warnings
from sklearn.linear_model import LinearRegression
import math
from sklearn import metrics
from sklearn import cluster
import chart_studio.plotly as py
import plotly.graph_objs as go
from plotly.offline import download_plotlyjs, init_notebook_mode, plot, iplot
import sklearn
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import RobustScaler
warnings.filterwarnings('ignore')
%matplotlib inline
```

Make sure to install the needed libraries

2 Another Dimension of Suicide Factors

2.1 Data Preprocessing: Reading, Integration, and Cleaning Data, etc

[471]:
```
df_hrs = pd.read_csv('human-rights-scores.csv')
df_hrv = pd.read_csv('human-rights-violations.csv')
df_pr = pd.read_csv('political-regime-updated2016.csv')
df_smr = pd.read_csv('suicide-mortality-rate.csv')
df_cp = pd.read_csv('TI-corruption-perception-index.csv')
df_fp = pd.read_csv('world-map-of-the-freedom-of-the-press-status.csv')
```

[472]:
```
df1 = pd.merge(df_hrs,df_pr, on=['Year','Code','Entity'])
df2 = pd.merge(df1,df_fp, on=['Year','Code','Entity'])
df3 = pd.merge(df2,df_cp, on=['Year','Code','Entity'])
df4 = pd.merge(df3,df_smr, on=['Year','Code','Entity'])
df4
```

[472]:
```
          Entity Code  Year  \
0    Afghanistan  AFG  2015
1        Albania  ALB  2015
2        Algeria  DZA  2015
3         Angola  AGO  2015
4      Argentina  ARG  2015
..           …  …   …
150    Venezuela  VEN  2015
151      Vietnam  VNM  2015
152        Yemen  YEM  2015
153       Zambia  ZMB  2015
154     Zimbabwe  ZWE  2015

     Human Rights Score (Schnakenberg & Fariss, 2014; Fariss, 2019)  \
0                                                -2.209410
1                                                 0.770309
2                                                 0.221587
3                                                -0.537383
4                                                 0.703494
..                                                      …
150                                              -1.066304
151                                              -0.367476
152                                              -2.045437
153                                              -0.111683
154                                              -0.850764

     Political Regime (OWID based on Polity IV and Wimmer & Min)  \
0                                                     -1
1                                                      9
2                                                      2
3                                                     -2
```

```
4                                                   9
..                                                  …
150                                                 4
151                                                -7
152                                                 0
153                                                 7
154                                                 4

     Freedom of the Press Status  \
0                           1500
1                           2500
2                           1500
3                           1500
4                           2500
..                           …
150                         1500
151                         1500
152                         1500
153                         1500
154                         1500

     Corruption Perception Index - Transparency International (2018)  \
0                                                   11
1                                                   36
2                                                   36
3                                                   15
4                                                   32
..                                                  …
150                                                 17
151                                                 31
152                                                 18
153                                                 38
154                                                 21

     Suicide mortality rate (per 100,000 population)
0                                                 4.8
1                                                 6.0
2                                                 3.2
3                                                 5.0
4                                                 8.8
..                                                …
150                                               3.8
151                                               7.2
152                                               8.6
153                                               6.1
154                                              10.6
```

```
[155 rows x 8 columns]
```

[473]:
```python
df4.to_csv("df.csv", index=False)
```

[474]:
```python
df_suicide = pd.read_csv('master.csv')
df_suicide
```

[474]:
```
          country  year     sex          age  suicides_no  population  \
0         Albania  1987    male  15-24 years           21      312900
1         Albania  1987    male  35-54 years           16      308000
2         Albania  1987  female  15-24 years           14      289700
3         Albania  1987    male    75+ years            1       21800
4         Albania  1987    male  25-34 years            9      274300
…           …    …     …          …           …          …
27815  Uzbekistan  2014  female  35-54 years          107     3620833
27816  Uzbekistan  2014  female    75+ years            9      348465
27817  Uzbekistan  2014    male   5-14 years           60     2762158
27818  Uzbekistan  2014  female   5-14 years           44     2631600
27819  Uzbekistan  2014  female  55-74 years           21     1438935

       suicides/100k pop    country-year  HDI for year  gdp_for_year ($)   \
0                   6.71     Albania1987           NaN      2,156,624,900
1                   5.19     Albania1987           NaN      2,156,624,900
2                   4.83     Albania1987           NaN      2,156,624,900
3                   4.59     Albania1987           NaN      2,156,624,900
4                   3.28     Albania1987           NaN      2,156,624,900
…                  …             …           …                 …
27815               2.96  Uzbekistan2014         0.675     63,067,077,179
27816               2.58  Uzbekistan2014         0.675     63,067,077,179
27817               2.17  Uzbekistan2014         0.675     63,067,077,179
27818               1.67  Uzbekistan2014         0.675     63,067,077,179
27819               1.46  Uzbekistan2014         0.675     63,067,077,179

       gdp_per_capita ($)       generation
0                     796     Generation X
1                     796           Silent
2                     796     Generation X
3                     796  G.I. Generation
4                     796          Boomers
…                    …             …
27815                2309     Generation X
27816                2309           Silent
27817                2309     Generation Z
27818                2309     Generation Z
27819                2309          Boomers

[27820 rows x 12 columns]
```

[475]:
```
df_suicide.drop('sex', axis=1, inplace=True)
df_suicide.drop('age', axis=1, inplace=True)
df_suicide.drop('country-year', axis=1, inplace=True)
df_suicide.drop('generation', axis=1, inplace=True)
df_suicide.drop('HDI for year', axis=1, inplace=True)
df_suicide.drop('gdp_per_capita ($)', axis=1, inplace=True)
df_suicide
```

[475]:
```
          country  year  suicides_no  population  suicides/100k pop  \
0         Albania  1987           21      312900               6.71
1         Albania  1987           16      308000               5.19
2         Albania  1987           14      289700               4.83
3         Albania  1987            1       21800               4.59
4         Albania  1987            9      274300               3.28
…           …   …          …          …                 …
27815  Uzbekistan  2014          107     3620833               2.96
27816  Uzbekistan  2014            9      348465               2.58
27817  Uzbekistan  2014           60     2762158               2.17
27818  Uzbekistan  2014           44     2631600               1.67
27819  Uzbekistan  2014           21     1438935               1.46

       gdp_for_year ($)
0          2,156,624,900
1          2,156,624,900
2          2,156,624,900
3          2,156,624,900
4          2,156,624,900
…                  …
27815     63,067,077,179
27816     63,067,077,179
27817     63,067,077,179
27818     63,067,077,179
27819     63,067,077,179

[27820 rows x 6 columns]
```

2.1.1 Handling Inconsistent Values

[476]:
```
array_1 = np.array(df1['Entity'].unique())
array_2 = np.array(df_suicide['country'].unique())
print(np.setdiff1d(array_2, array_1))
print(np.setdiff1d(array_1, array_2))
```

```
['Antigua and Barbuda' 'Aruba' 'Bahamas' 'Barbados' 'Belize'
 'Bosnia and Herzegovina' 'Cabo Verde' 'Czech Republic' 'Dominica'
 'Grenada' 'Iceland' 'Kiribati' 'Macau' 'Maldives' 'Malta' 'Puerto Rico'
 'Republic of Korea' 'Russian Federation' 'Saint Kitts and Nevis'
 'Saint Lucia' 'Saint Vincent and Grenadines' 'San Marino' 'Serbia'
```

```
 'Seychelles']
['Afghanistan' 'Algeria' 'Angola' 'Bangladesh' 'Benin' 'Bhutan' 'Bolivia'
 'Botswana' 'Burkina Faso' 'Burundi' 'Cambodia' 'Cameroon' 'Cape Verde'
 'Central African Republic' 'Chad' 'China' 'Comoros' 'Congo'
 "Cote d'Ivoire" 'Czechia' 'Democratic Republic of Congo' 'Djibouti'
 'Dominican Republic' 'Egypt' 'Equatorial Guinea' 'Eritrea' 'Eswatini'
 'Ethiopia' 'Gabon' 'Gambia' 'Ghana' 'Guinea' 'Guinea-Bissau' 'Haiti'
 'Honduras' 'India' 'Indonesia' 'Iran' 'Iraq' 'Jordan' 'Kenya' 'Laos'
 'Lebanon' 'Lesotho' 'Liberia' 'Libya' 'Madagascar' 'Malawi' 'Malaysia'
 'Mali' 'Mauritania' 'Moldova' 'Morocco' 'Mozambique' 'Myanmar' 'Namibia'
 'Nepal' 'Niger' 'Nigeria' 'North Korea' 'North Macedonia' 'Pakistan'
 'Papua New Guinea' 'Peru' 'Russia' 'Rwanda' 'Saudi Arabia' 'Senegal'
 'Sierra Leone' 'Solomon Islands' 'Somalia' 'South Korea' 'South Sudan'
 'Sudan' 'Syria' 'Taiwan' 'Tajikistan' 'Tanzania' 'Timor' 'Togo' 'Tunisia'
 'Uganda' 'Venezuela' 'Vietnam' 'Yemen' 'Zambia' 'Zimbabwe']
```

[477]:
```
df_suicide = df_suicide.replace(to_replace=r'Russian Federation',␣
↪value='Russia')
df_suicide = df_suicide.replace(to_replace=r'Republic of Korea', value='South␣
↪Korea')
```

[478]:
```
df_suicide
```

[478]:
```
          country  year  suicides_no  population  suicides/100k pop  \
0         Albania  1987           21      312900               6.71
1         Albania  1987           16      308000               5.19
2         Albania  1987           14      289700               4.83
3         Albania  1987            1       21800               4.59
4         Albania  1987            9      274300               3.28
…           …    …          …          …                  …
27815  Uzbekistan  2014          107     3620833               2.96
27816  Uzbekistan  2014            9      348465               2.58
27817  Uzbekistan  2014           60     2762158               2.17
27818  Uzbekistan  2014           44     2631600               1.67
27819  Uzbekistan  2014           21     1438935               1.46

       gdp_for_year ($)
0          2,156,624,900
1          2,156,624,900
2          2,156,624,900
3          2,156,624,900
4          2,156,624,900
…                    …
27815     63,067,077,179
27816     63,067,077,179
27817     63,067,077,179
27818     63,067,077,179
```

```
27819     63,067,077,179

[27820 rows x 6 columns]
```

[479]:
```
df_suicide.isna().sum()
```

[479]:
```
country               0
year                  0
suicides_no           0
population            0
suicides/100k pop     0
 gdp_for_year ($)     0
dtype: int64
```

2.1.2 Handling Missing Data and Data Transformation

[480]:
```
new_suicide = df_suicide.groupby(['country','year'])
new_suicide = new_suicide.sum(lambda x: x['year'].sum()).reset_index()
new_suicide['suicides/100k pop'] = (new_suicide.suicides_no / new_suicide.
 ↪population) * 100000
new_suicide.rename(columns={'country': 'Entity'}, inplace=True)
new_suicide.rename(columns={'year': 'Year'}, inplace=True)
new_suicide['suicides/100k pop'] = new_suicide['suicides/100k pop'].replace({0:
 ↪np.nan})
new_suicide
```

[480]:

	Entity	Year	suicides_no	population	suicides/100k pop
0	Albania	1987	73	2709600	2.694125
1	Albania	1988	63	2764300	2.279058
2	Albania	1989	68	2803100	2.425886
3	Albania	1992	47	2822500	1.665190
4	Albania	1993	73	2807300	2.600363
…	…	…	…	…	…
2316	Uzbekistan	2010	1464	25651783	5.707206
2317	Uzbekistan	2011	1640	25978049	6.313022
2318	Uzbekistan	2012	1835	26381830	6.955545
2319	Uzbekistan	2013	1950	26838924	7.265567
2320	Uzbekistan	2014	2095	27313507	7.670198

[2321 rows x 5 columns]

[481]:
```
new_suicide.isna().sum()
```

[481]:
```
Entity                0
Year                  0
suicides_no           0
population            0
```

```
suicides/100k pop    65
dtype: int64
```

[482]:
```
new_suicide['suicides/100k pop'] = new_suicide.groupby(['Entity'],
                                                        sort=False)['suicides/
 ↪100k pop'].apply(lambda x: x.fillna(x.mean()))
new_suicide
```

[482]:

	Entity	Year	suicides_no	population	suicides/100k pop
0	Albania	1987	73	2709600	2.694125
1	Albania	1988	63	2764300	2.279058
2	Albania	1989	68	2803100	2.425886
3	Albania	1992	47	2822500	1.665190
4	Albania	1993	73	2807300	2.600363
…	…	…	…	…	…
2316	Uzbekistan	2010	1464	25651783	5.707206
2317	Uzbekistan	2011	1640	25978049	6.313022
2318	Uzbekistan	2012	1835	26381830	6.955545
2319	Uzbekistan	2013	1950	26838924	7.265567
2320	Uzbekistan	2014	2095	27313507	7.670198

[2321 rows x 5 columns]

2.1.3 Data and Dimensionality Reduction

[483]:
```
dff1 = pd.merge(df_hrs,df_pr, on=['Year','Code','Entity'])
dff2 = pd.merge(dff1,df_fp, on=['Year','Code','Entity'])
dff3 = pd.merge(dff2,df_cp, on=['Year','Code','Entity'])
dff4 = pd.merge(dff3,new_suicide, on=['Year','Entity'])
dff4
```

[483]:

	Entity	Code	Year	\
0	Argentina	ARG	2012	
1	Argentina	ARG	2013	
2	Argentina	ARG	2014	
3	Argentina	ARG	2015	
4	Armenia	ARM	2012	
..	…	…	…	
247	Uruguay	URY	2014	
248	Uruguay	URY	2015	
249	Uzbekistan	UZB	2012	
250	Uzbekistan	UZB	2013	
251	Uzbekistan	UZB	2014	

	Human Rights Score (Schnakenberg & Fariss, 2014; Fariss, 2019)	\
0	0.641149	
1	0.623129	

```
2                                          0.567474
3                                          0.703494
4                                         -0.048950
..                                              …
247                                        2.679902
248                                        2.725541
249                                       -0.519791
250                                       -0.577395
251                                       -0.681276

     Political Regime (OWID based on Polity IV and Wimmer & Min)  \
0                                                             8
1                                                             8
2                                                             8
3                                                             9
4                                                             5
..                                                            …
247                                                          10
248                                                          10
249                                                          -9
250                                                          -9
251                                                          -9

     Freedom of the Press Status  \
0                           2500
1                           2500
2                           2500
3                           2500
4                           1500
..                           …
247                         3500
248                         3500
249                         1500
250                         1500
251                         1500

     Corruption Perception Index - Transparency International (2018)  \
0                                                            35
1                                                            34
2                                                            34
3                                                            32
4                                                            34
..                                                            …
247                                                          73
248                                                          74
249                                                          17
250                                                          17
```

```
251                                                 18

     suicides_no  population  suicides/100k pop
0           3248    38441778           8.449141
1           2987    38859125           7.686740
2           3231    39276555           8.226282
3           3073    39699624           7.740627
4             79     2814300           2.807092
..            …         …                 …
247          591     3177645          18.598679
248          630     3190795          19.744296
249         1835    26381830           6.955545
250         1950    26838924           7.265567
251         2095    27313507           7.670198

[252 rows x 10 columns]
```

[484]:
```
dfff1 = pd.merge(df_hrv,df_pr, on=['Year','Code','Entity'])
dfff2 = pd.merge(dfff1,df_fp, on=['Year','Code','Entity'])
dfff3 = pd.merge(dfff2,df_cp, on=['Year','Code','Entity'])
dfff4 = pd.merge(dfff3,new_suicide, on=['Year','Entity'])
dfff4
```

[484]:
```
         Entity Code  Year  \
0     Argentina  ARG  2012
1     Argentina  ARG  2013
2     Argentina  ARG  2014
3       Armenia  ARM  2012
4       Armenia  ARM  2013
..          …  …   …
198     Uruguay  URY  2013
199     Uruguay  URY  2014
200  Uzbekistan  UZB  2012
201  Uzbekistan  UZB  2013
202  Uzbekistan  UZB  2014

     Fund for Peace (Fragile States Index (Human Rights Dimension))  \
0                                                            3.8
1                                                            4.1
2                                                            4.4
3                                                            6.8
4                                                            6.8
..                                                           …
198                                                          2.3
199                                                          2.2
200                                                          9.1
201                                                          9.2
```

```
202                                                 9.3

     Political Regime (OWID based on Polity IV and Wimmer & Min)  \
0                                                     8
1                                                     8
2                                                     8
3                                                     5
4                                                     5
..                                                   …
198                                                  10
199                                                  10
200                                                  -9
201                                                  -9
202                                                  -9

     Freedom of the Press Status  \
0                           2500
1                           2500
2                           2500
3                           1500
4                           1500
..                           …
198                         3500
199                         3500
200                         1500
201                         1500
202                         1500

     Corruption Perception Index - Transparency International (2018)  \
0                                                    35
1                                                    34
2                                                    34
3                                                    34
4                                                    36
..                                                   …
198                                                  73
199                                                  73
200                                                  17
201                                                  17
202                                                  18

     suicides_no  population  suicides/100k pop
0           3248    38441778           8.449141
1           2987    38859125           7.686740
2           3231    39276555           8.226282
3             79     2814300           2.807092
4             67     2810664           2.383778
```

```
..            …          …                 …
198          546     3164670          17.252984
199          591     3177645          18.598679
200         1835    26381830           6.955545
201         1950    26838924           7.265567
202         2095    27313507           7.670198

[203 rows x 10 columns]
```

[485]:
```
dff4.rename(columns={'Human Rights Score (Schnakenberg & Fariss, 2014; Fariss,␣
↪2019)': 'Human Rights Score'}, inplace=True)
dff4.rename(columns={'Political Regime (OWID based on Polity IV and Wimmer &␣
↪Min)': 'Political Regime Score'}, inplace=True)
dff4.rename(columns={'Corruption Perception Index - Transparency International␣
↪(2018)': 'Corruption Perception Index'}, inplace=True)
dff4.drop('population', axis=1, inplace=True)
dff4.drop('Code', axis=1, inplace=True)
#dff4.drop('suicides_no', axis=1, inplace=True)
dfff4.rename(columns={'Fund for Peace (Fragile States Index (Human Rights␣
↪Dimension))': 'Human Rights Violation'}, inplace=True)
dfff4.rename(columns={'Political Regime (OWID based on Polity IV and Wimmer &␣
↪Min)': 'Political Regime Score'}, inplace=True)
dfff4.rename(columns={'Corruption Perception Index - Transparency International␣
↪(2018)': 'Corruption Perception Index'}, inplace=True)
dfff4.drop('Code', axis=1, inplace=True)
dfff4.drop('population', axis=1, inplace=True)
#dfff4.drop('suicides_no', axis=1, inplace=True)
```

[486]:
```
dff4.to_csv("df1.csv", index=False)
dfff4.to_csv("df2.csv", index=False)
```

[487]:
```
dff4
```

[487]:
```
         Entity  Year  Human Rights Score  Political Regime Score  \
0     Argentina  2012            0.641149                       8
1     Argentina  2013            0.623129                       8
2     Argentina  2014            0.567474                       8
3     Argentina  2015            0.703494                       9
4       Armenia  2012           -0.048950                       5
..          …   …                  …                       …
247     Uruguay  2014            2.679902                      10
248     Uruguay  2015            2.725541                      10
249  Uzbekistan  2012           -0.519791                      -9
250  Uzbekistan  2013           -0.577395                      -9
251  Uzbekistan  2014           -0.681276                      -9

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
```

```
0                          2500                          35         3248
1                          2500                          34         2987
2                          2500                          34         3231
3                          2500                          32         3073
4                          1500                          34           79
..                          …                           …          …
247                        3500                          73          591
248                        3500                          74          630
249                        1500                          17         1835
250                        1500                          17         1950
251                        1500                          18         2095

     suicides/100k pop
0             8.449141
1             7.686740
2             8.226282
3             7.740627
4             2.807092
..                 …
247          18.598679
248          19.744296
249           6.955545
250           7.265567
251           7.670198

[252 rows x 8 columns]
```

[488]:
```
dfff4
```

[488]:
```
         Entity  Year  Human Rights Violation  Political Regime Score  \
0     Argentina  2012                     3.8                       8
1     Argentina  2013                     4.1                       8
2     Argentina  2014                     4.4                       8
3       Armenia  2012                     6.8                       5
4       Armenia  2013                     6.8                       5
..          …   …                       …                       …
198     Uruguay  2013                     2.3                      10
199     Uruguay  2014                     2.2                      10
200  Uzbekistan  2012                     9.1                      -9
201  Uzbekistan  2013                     9.2                      -9
202  Uzbekistan  2014                     9.3                      -9

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                           2500                           35         3248
1                           2500                           34         2987
2                           2500                           34         3231
3                           1500                           34           79
```

```
4                           1500                            36           67
..                           …                           …          …
198                         3500                            73          546
199                         3500                            73          591
200                         1500                            17         1835
201                         1500                            17         1950
202                         1500                            18         2095

     suicides/100k pop
0             8.449141
1             7.686740
2             8.226282
3             2.807092
4             2.383778
..                 …
198          17.252984
199          18.598679
200           6.955545
201           7.265567
202           7.670198

[203 rows x 8 columns]
```

[489]:
```
df4.rename(columns={'Human Rights Score (Schnakenberg & Fariss, 2014; Fariss,␣
↪2019)': 'Human Rights Score'}, inplace=True)
df4.rename(columns={'Political Regime (OWID based on Polity IV and Wimmer &␣
↪Min)': 'Political Regime Score'}, inplace=True)
df4.rename(columns={'Corruption Perception Index - Transparency International␣
↪(2018)': 'Corruption Perception Index'}, inplace=True)
df4.rename(columns={'Suicide mortality rate (per 100,000 population)':␣
↪'suicides/100k pop'}, inplace=True)
df4.drop('Code', axis=1, inplace=True)
Data_2015 = df4
Data_HRS = dff4
Data_HRV = dfff4
```

2.1.4 Normalization and final touch before the analysis

[490]:
```
Data_2015.describe()
```

[490]:
```
          Year  Human Rights Score  Political Regime Score  \
count   155.0          155.000000              155.000000
mean   2015.0            0.365058                4.451613
std       0.0            1.472837                5.990462
min    2015.0           -2.550781              -10.000000
25%    2015.0           -0.534317                0.000000
50%    2015.0            0.261105                7.000000
```

```
75%    2015.0            1.332057                 9.000000
max    2015.0            5.139970                10.000000

       Freedom of the Press Status  Corruption Perception Index  \
count                   155.000000                   155.000000
mean                   2383.870968                    42.496774
std                     772.538729                    20.182085
min                    1500.000000                     8.000000
25%                    1500.000000                    28.000000
50%                    2500.000000                    37.000000
75%                    2500.000000                    53.500000
max                    3500.000000                    91.000000

       suicides/100k pop
count         155.000000
mean            9.870968
std             6.325079
min             2.100000
25%             5.000000
50%             8.500000
75%            12.650000
max            34.900000
```

[491]:
```
Data_HRS.describe()
```

[491]:
```
              Year  Human Rights Score  Political Regime Score  \
count   252.000000          252.000000              252.000000
mean   2013.408730            1.121710                6.619048
std       1.091256            1.460847                5.712720
min    2012.000000           -1.449019              -10.000000
25%    2012.000000            0.036572                7.000000
50%    2013.000000            1.248456                9.000000
75%    2014.000000            2.127065               10.000000
max    2015.000000            5.139970               10.000000

       Freedom of the Press Status  Corruption Perception Index   suicides_no  \
count                   252.000000                   252.000000    252.000000
mean                   2738.095238                    54.865079   3442.456349
std                     797.691604                    20.587914   7109.743203
min                    1500.000000                    17.000000      5.000000
25%                    2500.000000                    37.000000    381.250000
50%                    2500.000000                    53.500000    891.500000
75%                    3500.000000                    73.000000   2997.000000
max                    3500.000000                    92.000000  44189.000000

       suicides/100k pop
count         252.000000
```

```
mean           11.642763
std             7.363635
min             0.192779
25%             5.603576
50%            11.014161
75%            16.189341
max            38.658504
```

[492]:
```
Data_HRV.describe()
```

[492]:
```
              Year  Human Rights Violation  Political Regime Score  \
count   203.000000              203.000000              203.000000
mean   2012.995074                4.258621                6.546798
std       0.817492                2.196681                5.817482
min    2012.000000                1.000000              -10.000000
25%    2012.000000                2.400000                7.000000
50%    2013.000000                3.700000                9.000000
75%    2014.000000                6.100000               10.000000
max    2014.000000                9.300000               10.000000

       Freedom of the Press Status  Corruption Perception Index   suicides_no  \
count                   203.000000                   203.000000    203.000000
mean                   2736.453202                    54.211823   3301.571429
std                     791.821582                    20.379796   6916.016750
min                    1500.000000                    17.000000      5.000000
25%                    2500.000000                    37.000000    380.500000
50%                    2500.000000                    52.000000    873.000000
75%                    3500.000000                    72.000000   2594.000000
max                    3500.000000                    92.000000  42769.000000

       suicides/100k pop
count         203.000000
mean           11.778888
std             7.480819
min             0.192779
25%             5.612273
50%            11.086206
75%            16.877339
max            38.658504
```

[493]:
```
Data_2015['Human Rights Score'] = preprocessing.minmax_scale(Data_2015['Human␣
↪Rights Score'], feature_range=(0, 1))
Data_2015['Political Regime Score'] = preprocessing.
↪minmax_scale(Data_2015['Political Regime Score'], feature_range=(0, 1))
Data_2015['Freedom of the Press Status'] = preprocessing.
↪minmax_scale(Data_2015['Freedom of the Press Status'], feature_range=(0, 1))
```

```
Data_2015['Corruption Perception Index'] = preprocessing.
↪minmax_scale(Data_2015['Corruption Perception Index'], feature_range=(0, 1))


Data_HRS['Human Rights Score'] = preprocessing.minmax_scale(Data_HRS['Human␣
↪Rights Score'], feature_range=(0, 1))
Data_HRS['Political Regime Score'] = preprocessing.
↪minmax_scale(Data_HRS['Political Regime Score'], feature_range=(0, 1))
Data_HRS['Freedom of the Press Status'] = preprocessing.
↪minmax_scale(Data_HRS['Freedom of the Press Status'], feature_range=(0, 1))
Data_HRS['Corruption Perception Index'] = preprocessing.
↪minmax_scale(Data_HRS['Corruption Perception Index'], feature_range=(0, 1))

Data_HRV['Human Rights Violation'] = preprocessing.minmax_scale(Data_HRV['Human␣
↪Rights Violation'], feature_range=(0, 1))
Data_HRV['Political Regime Score'] = preprocessing.
↪minmax_scale(Data_HRV['Political Regime Score'], feature_range=(0, 1))
Data_HRV['Freedom of the Press Status'] = preprocessing.
↪minmax_scale(Data_HRV['Freedom of the Press Status'], feature_range=(0, 1))
Data_HRV['Corruption Perception Index'] = preprocessing.
↪minmax_scale(Data_HRV['Corruption Perception Index'], feature_range=(0, 1))
```

[494]:
```
Data_2015
```

[494]:
```
          Entity  Year  Human Rights Score  Political Regime Score  \
0    Afghanistan  2015            0.044387                    0.45
1        Albania  2015            0.431829                    0.95
2        Algeria  2015            0.360481                    0.60
3         Angola  2015            0.261795                    0.40
4      Argentina  2015            0.423141                    0.95
..           …   …                  …                       …
150    Venezuela  2015            0.193021                    0.70
151      Vietnam  2015            0.283887                    0.15
152        Yemen  2015            0.065708                    0.50
153       Zambia  2015            0.317147                    0.85
154     Zimbabwe  2015            0.221047                    0.70

     Freedom of the Press Status  Corruption Perception Index  \
0                            0.0                     0.036145
1                            0.5                     0.337349
2                            0.0                     0.337349
3                            0.0                     0.084337
4                            0.5                     0.289157
..                           …                        …
150                          0.0                     0.108434
151                          0.0                     0.277108
```

```
152                          0.0                      0.120482
153                          0.0                      0.361446
154                          0.0                      0.156627

     suicides/100k pop
0                  4.8
1                  6.0
2                  3.2
3                  5.0
4                  8.8
..                 …
150                3.8
151                7.2
152                8.6
153                6.1
154               10.6

[155 rows x 7 columns]
```

[495]:
```
Data_HRS
```

[495]:
```
         Entity  Year  Human Rights Score  Political Regime Score  \
0     Argentina  2012            0.317221                    0.90
1     Argentina  2013            0.314486                    0.90
2     Argentina  2014            0.306040                    0.90
3     Argentina  2015            0.326683                    0.95
4       Armenia  2012            0.212486                    0.75
..          …   …                  …                       …
247     Uruguay  2014            0.626640                    1.00
248     Uruguay  2015            0.633566                    1.00
249  Uzbekistan  2012            0.141027                    0.05
250  Uzbekistan  2013            0.132285                    0.05
251  Uzbekistan  2014            0.116519                    0.05

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                            0.5                     0.240000         3248
1                            0.5                     0.226667         2987
2                            0.5                     0.226667         3231
3                            0.5                     0.200000         3073
4                            0.0                     0.226667           79
..                           …                        …          …
247                          1.0                     0.746667          591
248                          1.0                     0.760000          630
249                          0.0                     0.000000         1835
250                          0.0                     0.000000         1950
251                          0.0                     0.013333         2095
```

```
     suicides/100k pop
0             8.449141
1             7.686740
2             8.226282
3             7.740627
4             2.807092
..                 …
247          18.598679
248          19.744296
249           6.955545
250           7.265567
251           7.670198

[252 rows x 8 columns]
```

[496]:
```
Data_HRV
```

[496]:
```
         Entity  Year  Human Rights Violation  Political Regime Score  \
0     Argentina  2012                0.337349                    0.90
1     Argentina  2013                0.373494                    0.90
2     Argentina  2014                0.409639                    0.90
3       Armenia  2012                0.698795                    0.75
4       Armenia  2013                0.698795                    0.75
..          …   …                     …                       …
198     Uruguay  2013                0.156627                    1.00
199     Uruguay  2014                0.144578                    1.00
200  Uzbekistan  2012                0.975904                    0.05
201  Uzbekistan  2013                0.987952                    0.05
202  Uzbekistan  2014                1.000000                    0.05

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                            0.5                     0.240000         3248
1                            0.5                     0.226667         2987
2                            0.5                     0.226667         3231
3                            0.0                     0.226667           79
4                            0.0                     0.253333           67
..                           …                        …          …
198                          1.0                     0.746667          546
199                          1.0                     0.746667          591
200                          0.0                     0.000000         1835
201                          0.0                     0.000000         1950
202                          0.0                     0.013333         2095

     suicides/100k pop
0             8.449141
1             7.686740
2             8.226282
```

```
3             2.807092
4             2.383778
..                 …
198          17.252984
199          18.598679
200           6.955545
201           7.265567
202           7.670198

[203 rows x 8 columns]
```

checking for NaN values

[497]:
```
Data_2015.isna().sum()
```

[497]:
```
Entity                        0
Year                          0
Human Rights Score            0
Political Regime Score        0
Freedom of the Press Status   0
Corruption Perception Index   0
suicides/100k pop             0
dtype: int64
```

[498]:
```
Data_HRS.isna().sum()
```

[498]:
```
Entity                        0
Year                          0
Human Rights Score            0
Political Regime Score        0
Freedom of the Press Status   0
Corruption Perception Index   0
suicides_no                   0
suicides/100k pop             0
dtype: int64
```

[499]:
```
Data_HRV.isna().sum()
```

[499]:
```
Entity                        0
Year                          0
Human Rights Violation        0
Political Regime Score        0
Freedom of the Press Status   0
Corruption Perception Index   0
suicides_no                   0
suicides/100k pop             0
dtype: int64
```

checking for uniqueness

[500]:
```
Data_HRS['Year'].unique()
```

[500]:
```
array([2012, 2013, 2014, 2015], dtype=int64)
```

[501]:
```
Data_HRV['Year'].unique()
```

[501]:
```
array([2012, 2013, 2014], dtype=int64)
```

[502]:
```
Data_2015['Entity'].nunique()
```

[502]:
```
155
```

[503]:
```
Data_HRS['Entity'].nunique()
```

[503]:
```
69
```

[504]:
```
Data_HRV['Entity'].nunique()
```

[504]:
```
70
```

A function that returns min and max of country regading a giving attribute and dataset

[505]:
```
def minmax(d, n):
    lowest = d[n].idxmin()
    highest = d[n].idxmax()
    high = pd.DataFrame(d.loc[highest,:])
    low = pd.DataFrame(d.loc[lowest,:])
    return pd.concat([high,low],axis = 1)
```

[506]:
```
minmax(Data_HRV, 'suicides/100k pop')
```

[506]:

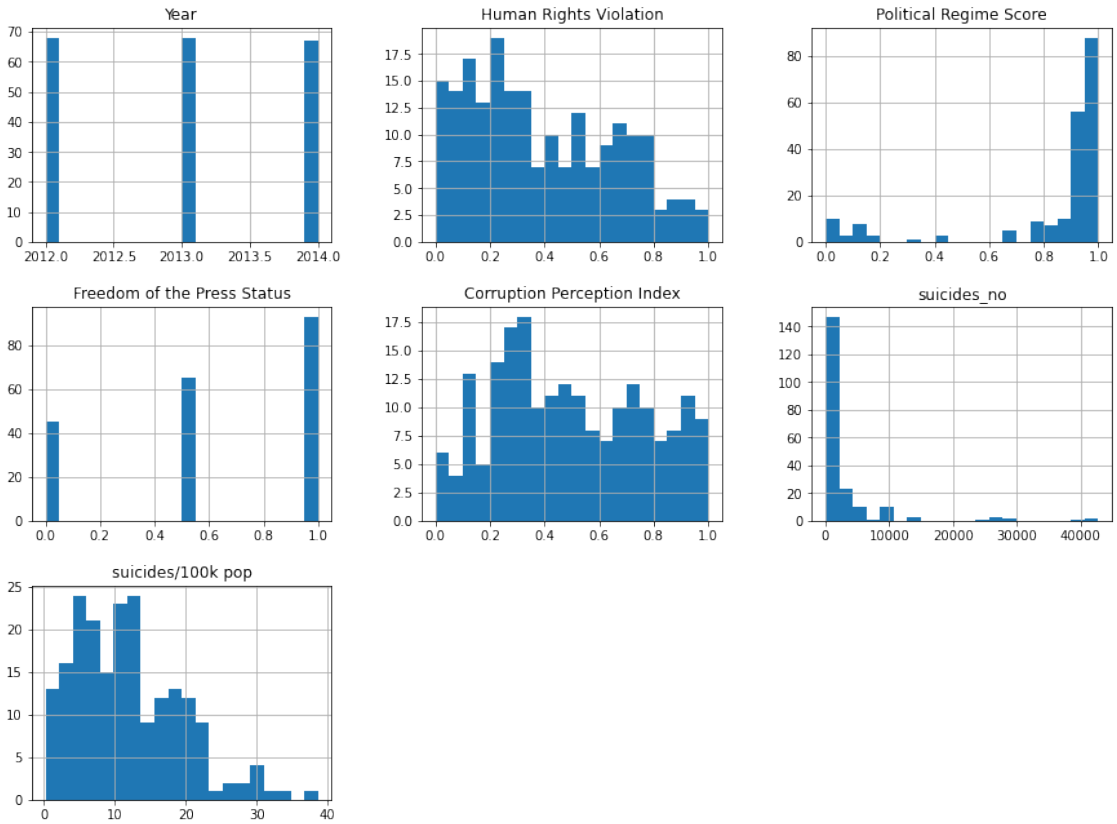
	106	128
Entity	Lithuania	Oman
Year	2013	2014
Human Rights Violation	0.228916	0.771084
Political Regime Score	1	0.1
Freedom of the Press Status	1	0
Corruption Perception Index	0.533333	0.373333
suicides_no	1085	7
suicides/100k pop	38.6585	0.192779

2.2 Data Analysis and Statistical Visualization

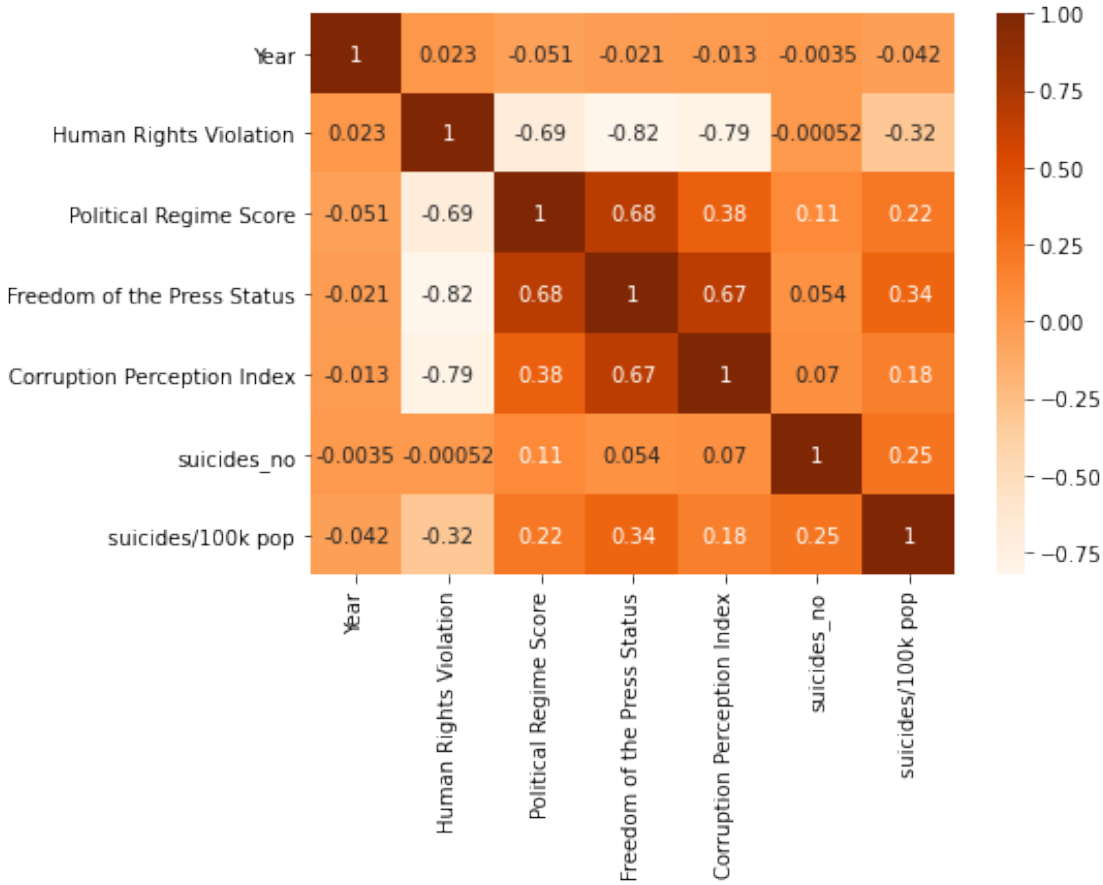
2.2.1 Data Containing 3 years for 70 countries

[507]:
```
Data_HRV.hist(bins = 20,figsize = (15,11))
```

[507]:
```
array([[<AxesSubplot:title={'center':'Year'}>,
        <AxesSubplot:title={'center':'Human Rights Violation'}>,
        <AxesSubplot:title={'center':'Political Regime Score'}>],
       [<AxesSubplot:title={'center':'Freedom of the Press Status'}>,
        <AxesSubplot:title={'center':'Corruption Perception Index'}>,
        <AxesSubplot:title={'center':'suicides_no'}>],
       [<AxesSubplot:title={'center':'suicides/100k pop'}>,
        <AxesSubplot:>, <AxesSubplot:>]], dtype=object)
```

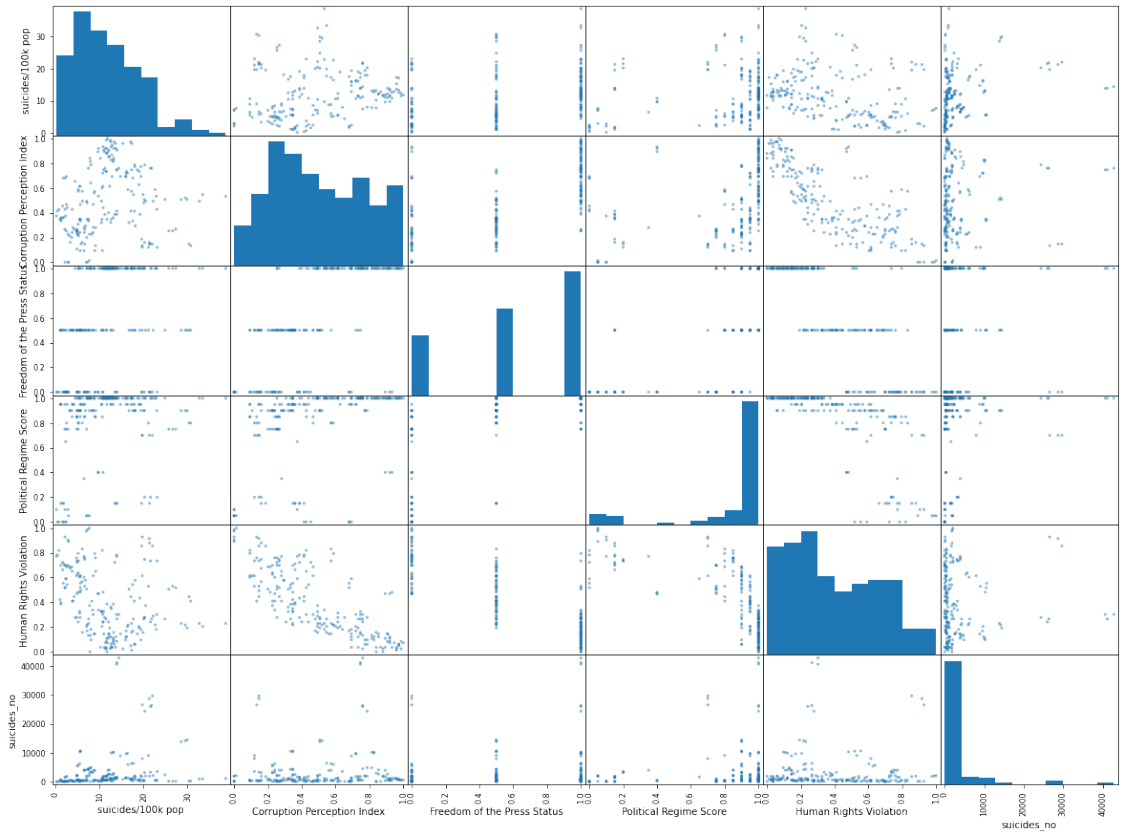


[508]:
```
plt.figure(figsize=(7,5))
sns.heatmap(Data_HRV.corr(), annot=True, cmap='Oranges')
plt.show()
```



[509]:
```
plt.figure(figsize=(20,15))
attributes = ['suicides/100k pop', 'Corruption Perception Index', 'Freedom of␣
↪the Press Status','Political Regime Score',
              'Human Rights Violation', 'suicides_no']
scatter_matrix(Data_HRV[attributes], figsize=(20,15))
plt.show()
```

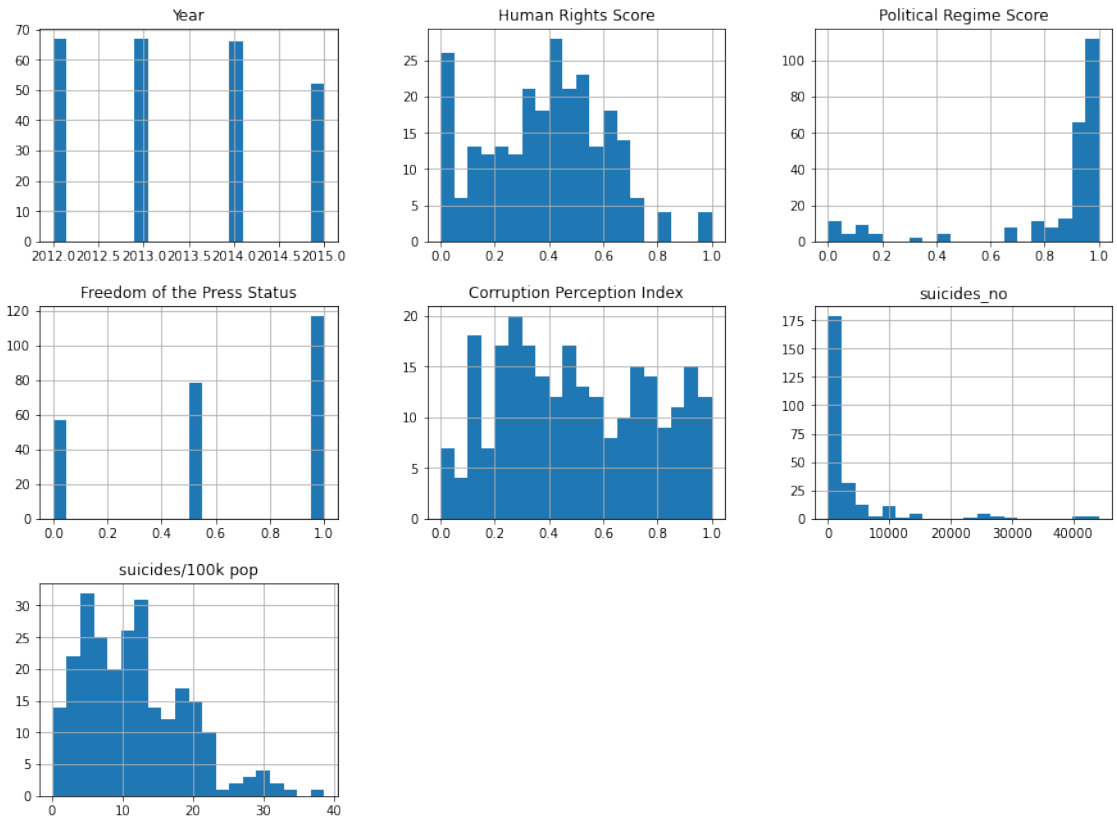
```
<Figure size 1440x1080 with 0 Axes>
```



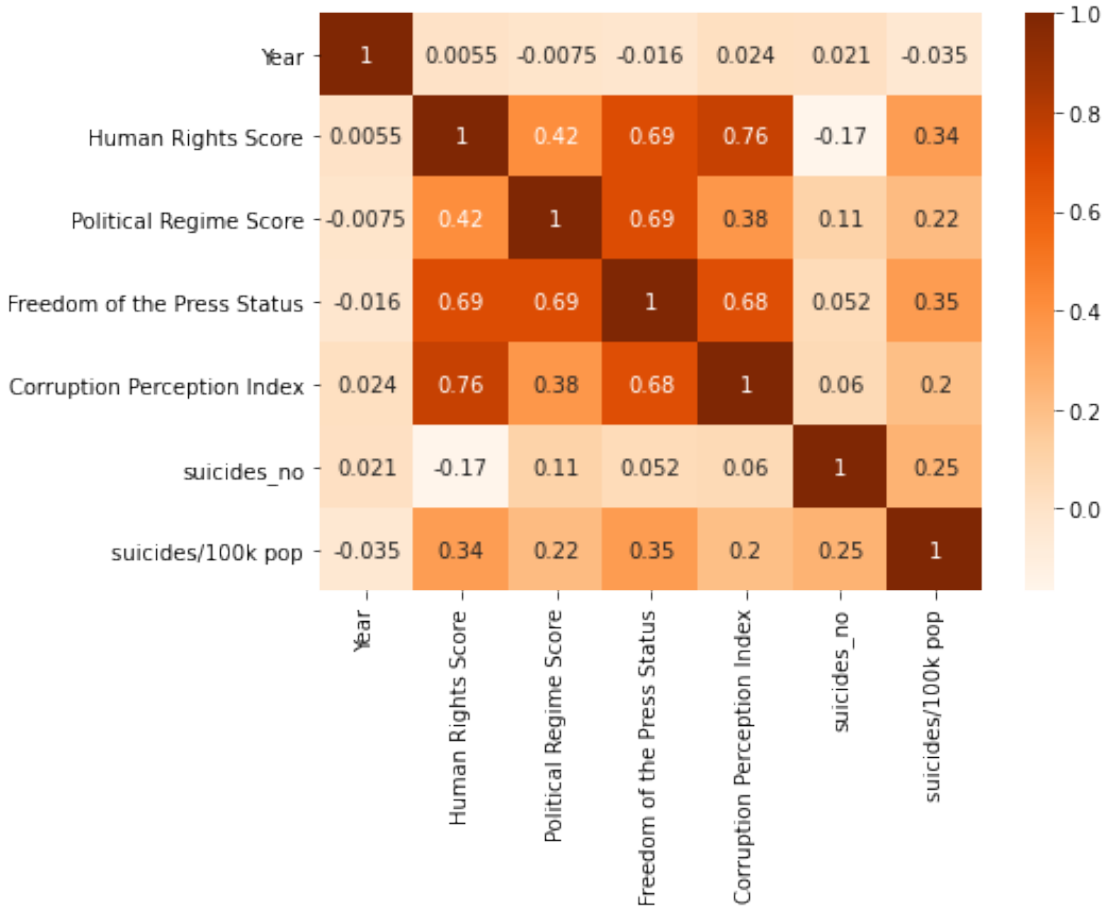
2.2.2 Data Containing 4 years for 70 countries

[510]:
```
Data_HRS.hist(bins = 20,figsize = (15,11))
```

[510]:
```
array([[<AxesSubplot:title={'center':'Year'}>,
        <AxesSubplot:title={'center':'Human Rights Score'}>,
        <AxesSubplot:title={'center':'Political Regime Score'}>],
       [<AxesSubplot:title={'center':'Freedom of the Press Status'}>,
        <AxesSubplot:title={'center':'Corruption Perception Index'}>,
        <AxesSubplot:title={'center':'suicides_no'}>],
       [<AxesSubplot:title={'center':'suicides/100k pop'}>,
        <AxesSubplot:>, <AxesSubplot:>]], dtype=object)
```

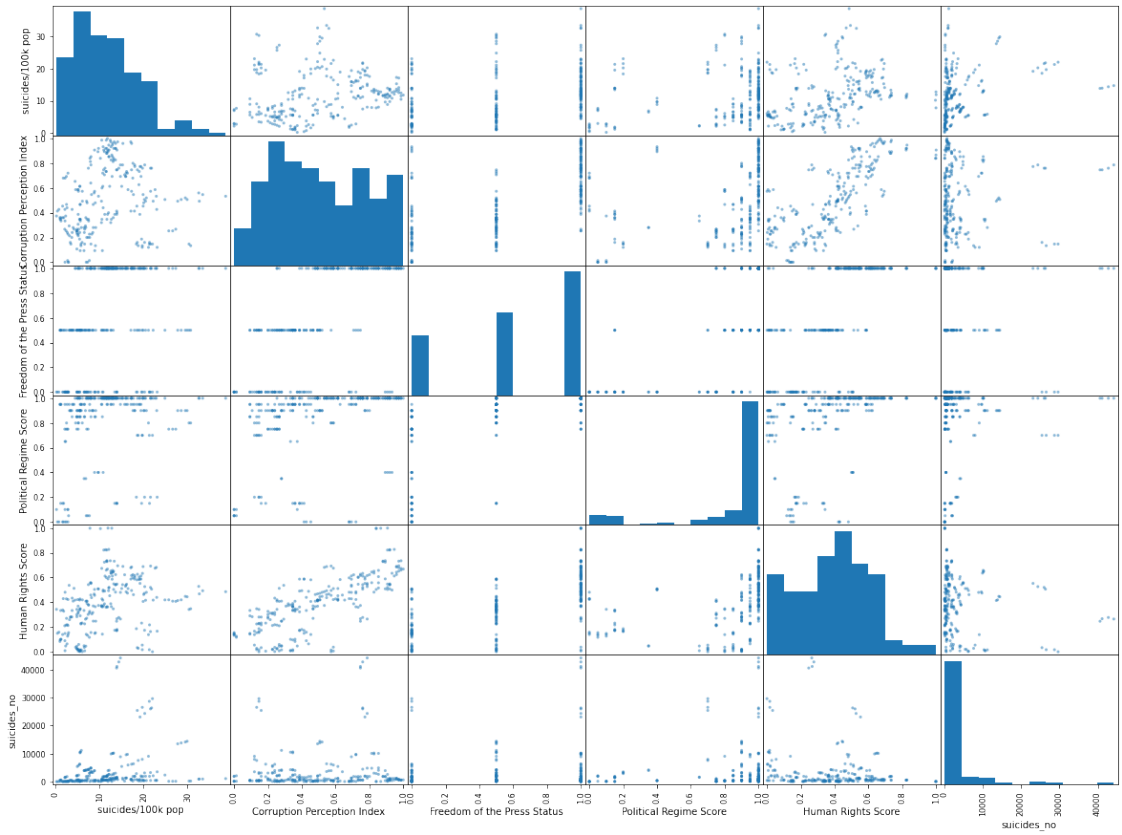


[511]:
```
plt.figure(figsize=(7,5))
sns.heatmap(Data_HRS.corr(), annot=True, cmap='Oranges')
plt.show()
```



[512]:
```
plt.figure(figsize=(20,15))
attributes = ['suicides/100k pop', 'Corruption Perception Index', 'Freedom of␣
↪the Press Status','Political Regime Score',
              'Human Rights Score', 'suicides_no']
scatter_matrix(Data_HRS[attributes], figsize=(20,15))
plt.show()
```

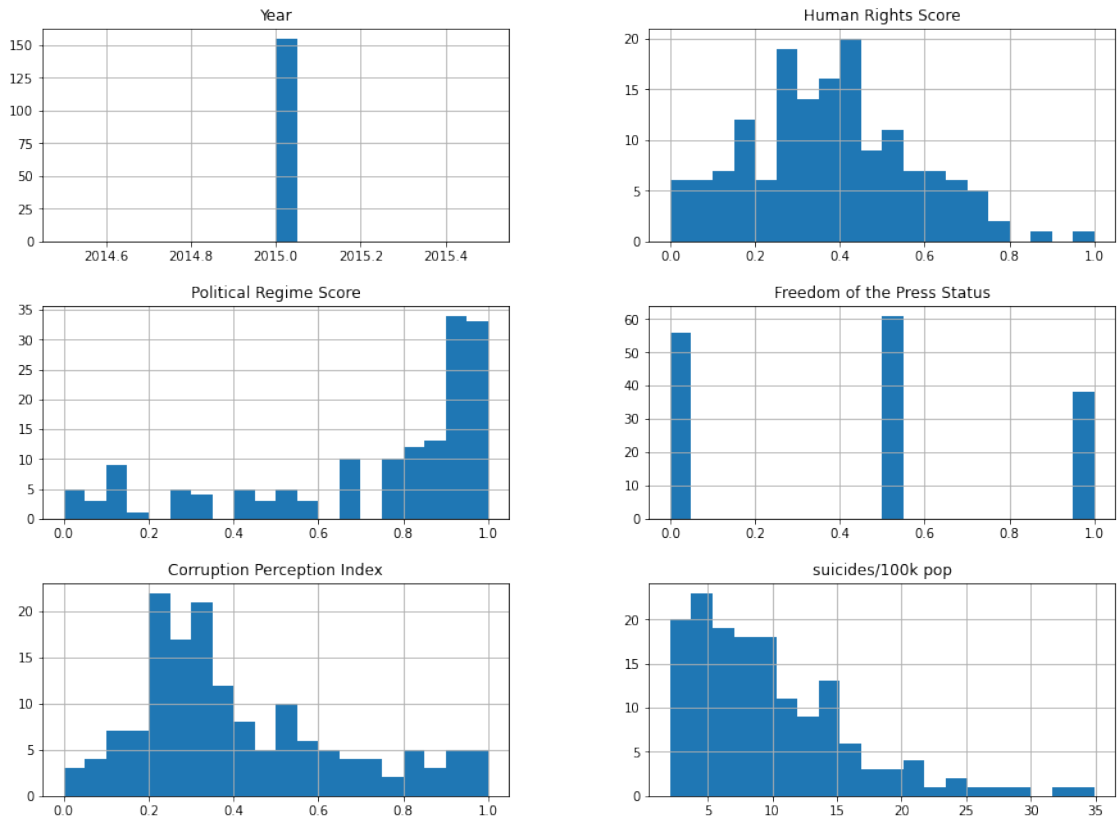
```
<Figure size 1440x1080 with 0 Axes>
```



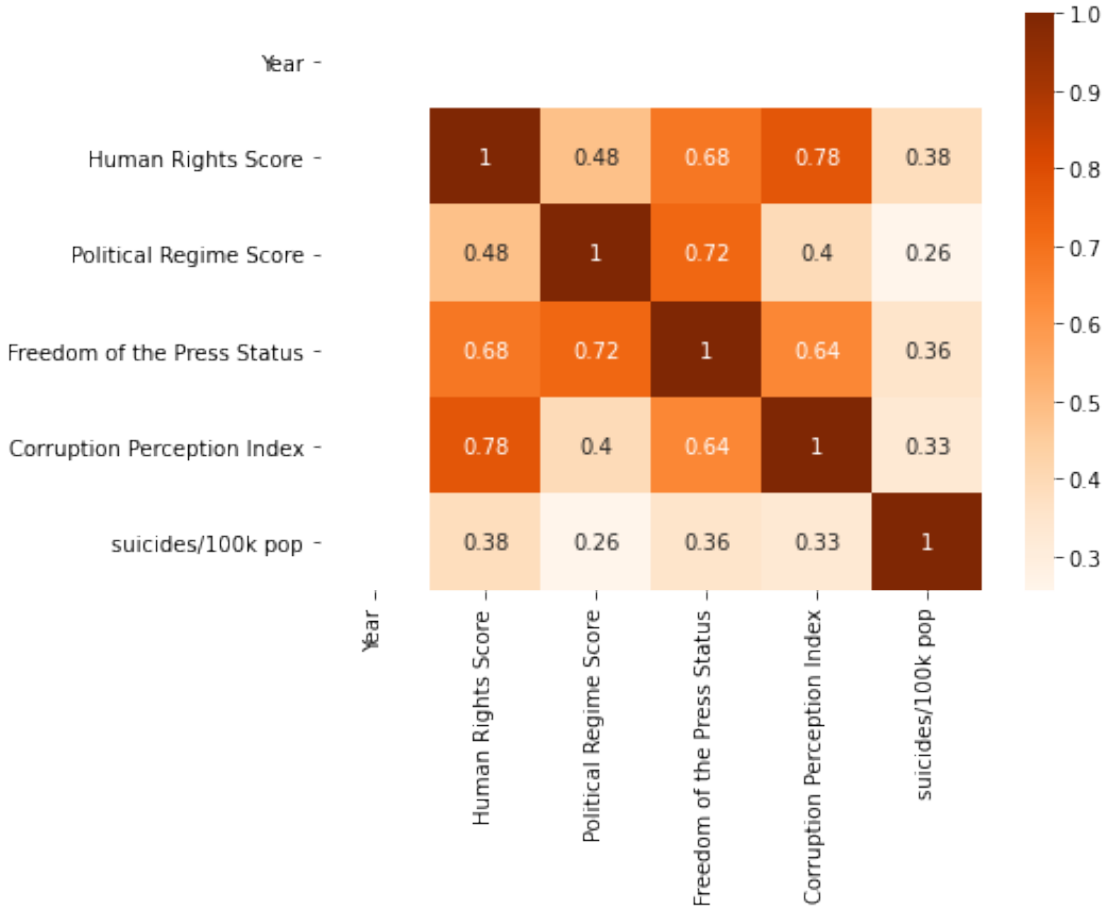
2.2.3 Data Containing 1 year for 155 countries

[513]:
```
Data_2015.hist(bins = 20,figsize = (15,11))
```

[513]:
```
array([[<AxesSubplot:title={'center':'Year'}>,
        <AxesSubplot:title={'center':'Human Rights Score'}>],
       [<AxesSubplot:title={'center':'Political Regime Score'}>,
        <AxesSubplot:title={'center':'Freedom of the Press Status'}>],
       [<AxesSubplot:title={'center':'Corruption Perception Index'}>,
        <AxesSubplot:title={'center':'suicides/100k pop'}>]], dtype=object)
```

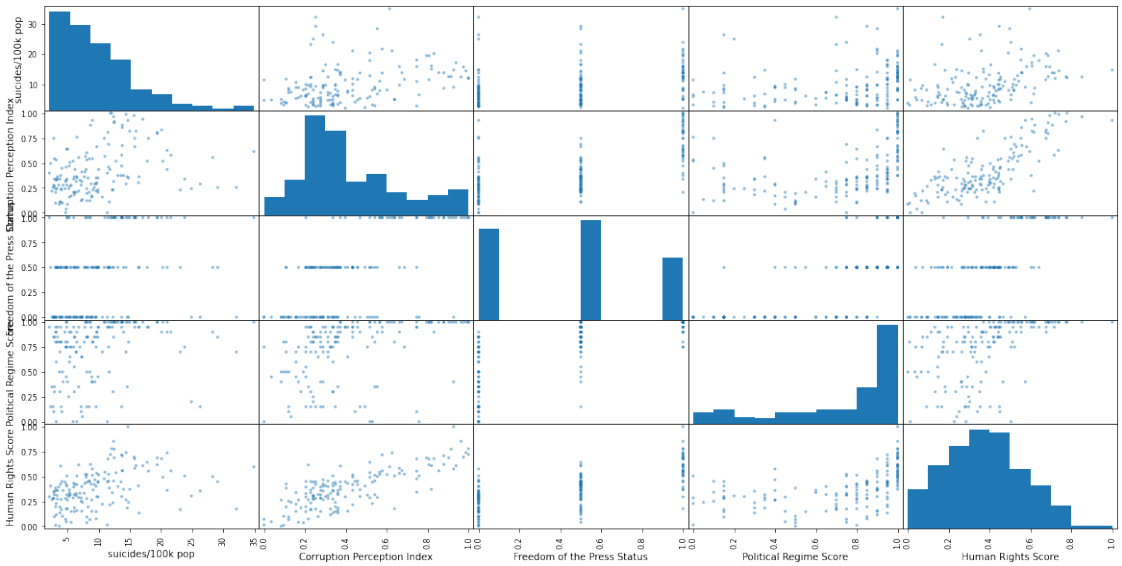


[514]:
```
plt.figure(figsize=(7,5))
sns.heatmap(Data_2015.corr(), annot=True, cmap='Oranges')
plt.show()
```



[515]:
```
plt.figure(figsize=(20,10))
attributes = ['suicides/100k pop', 'Corruption Perception Index', 'Freedom of␣
↪the Press Status','Political Regime Score',
              'Human Rights Score']
scatter_matrix(Data_2015[attributes], figsize=(20,10))
plt.show()
```

```
<Figure size 1440x720 with 0 Axes>
```



2.3 Data Mining

[516]:
```
suicides_df = pd.read_csv('df1.csv')
```

[517]:
```
suicides_df.head(5)
```

[517]:
```
       Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.641149                       8
1  Argentina  2013            0.623129                       8
2  Argentina  2014            0.567474                       8
3  Argentina  2015            0.703494                       9
4    Armenia  2012           -0.048950                       5

   Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                         2500                           35         3248
1                         2500                           34         2987
2                         2500                           34         3231
3                         2500                           32         3073
4                         1500                           34           79

   suicides/100k pop
0           8.449141
1           7.686740
2           8.226282
3           7.740627
4           2.807092
```

[518]:
```
suicides_df.drop('suicides_no', axis=1, inplace=True)
```

[519]:
```
suicides_df.head(2)
```

[519]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.641149                       8
1  Argentina  2013            0.623129                       8

   Freedom of the Press Status  Corruption Perception Index  suicides/100k pop
0                         2500                           35           8.449141
1                         2500                           34           7.686740
```

[520]:
```
suicides_df['Human Rights Score'] = suicides_df['Human Rights Score'] + (1.
↪4490186369999998)
suicides_df['Political Regime Score'] = suicides_df['Political Regime Score'] +␣
↪(10)
```

[521]:
```
print("The min value in the HRS column is", suicides_df['Human Rights Score'].
↪min())
print("The min value in the Political Regime column is", suicides_df['Political␣
↪Regime Score'].min())
```

```
The min value in the HRS column is 0.0
The min value in the Political Regime column is 0
```

[522]:
```
normalized_df = suicides_df.copy()
```

[523]:
```
normalized_df.head(5)
```

[523]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            2.090168                      18
1  Argentina  2013            2.072147                      18
2  Argentina  2014            2.016493                      18
3  Argentina  2015            2.152512                      19
4    Armenia  2012            1.400068                      15

   Freedom of the Press Status  Corruption Perception Index  suicides/100k pop
0                         2500                           35           8.449141
1                         2500                           34           7.686740
2                         2500                           34           8.226282
3                         2500                           32           7.740627
4                         1500                           34           2.807092
```

[524]:
```
normalized_df['Human Rights Score'] /= normalized_df['Human Rights Score'].max()
normalized_df['Political Regime Score'] /= normalized_df['Political Regime␣
↪Score'].max()
normalized_df['Freedom of the Press Status'] /= normalized_df['Freedom of the␣
↪Press Status'].max()
normalized_df['Corruption Perception Index'] /= normalized_df['Corruption␣
↪Perception Index'].max()
```

[525]:
```
normalized_df.head(5)
```

[525]:

	Entity	Year	Human Rights Score	Political Regime Score	\
0	Argentina	2012	0.317221	0.90	
1	Argentina	2013	0.314486	0.90	
2	Argentina	2014	0.306040	0.90	
3	Argentina	2015	0.326683	0.95	
4	Armenia	2012	0.212486	0.75	

	Freedom of the Press Status	Corruption Perception Index	suicides/100k pop
0	0.714286	0.380435	8.449141
1	0.714286	0.369565	7.686740
2	0.714286	0.369565	8.226282
3	0.714286	0.347826	7.740627
4	0.428571	0.369565	2.807092

[526]:
```
features = ['Human Rights Score', 'Political Regime Score', 'Freedom of the␣
↪Press Status',
            'Corruption Perception Index']
X = normalized_df[features]
Y = normalized_df['suicides/100k pop']
```

2.3.1 Linear Regression

[527]:
```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.3,␣
↪random_state=101)
```

[528]:
```
lm = LinearRegression(fit_intercept=True, normalize=False, copy_X=True,␣
↪n_jobs=None)
```

[529]:
```
lm.fit(X_train, Y_train)
```

[529]:
```
LinearRegression()
```

[530]:
```
print("The intercept of the regression line is", lm.intercept_)
```

```
The intercept of the regression line is 4.871561001626349
```

[531]:
```
print("The coefficients are", lm.coef_)
```

```
The coefficients are [ 9.67955702 -1.17040581 11.94523384 -9.16180206]
```

[532]:
```
coefficients_df = pd.DataFrame(lm.coef_, features, columns=['Coefficients'])
```

[533]:
```
coefficients_df
```

[533]:

	Coefficients
Human Rights Score	9.679557
Political Regime Score	-1.170406

```
Freedom of the Press Status     11.945234
Corruption Perception Index     -9.161802
```

How to read the coefficient dataframe:

If all other features were held constant, a one unit increase in the Human Rights Score leads to an increase of 9.679557 in the suicide mortality rate per 100k people.

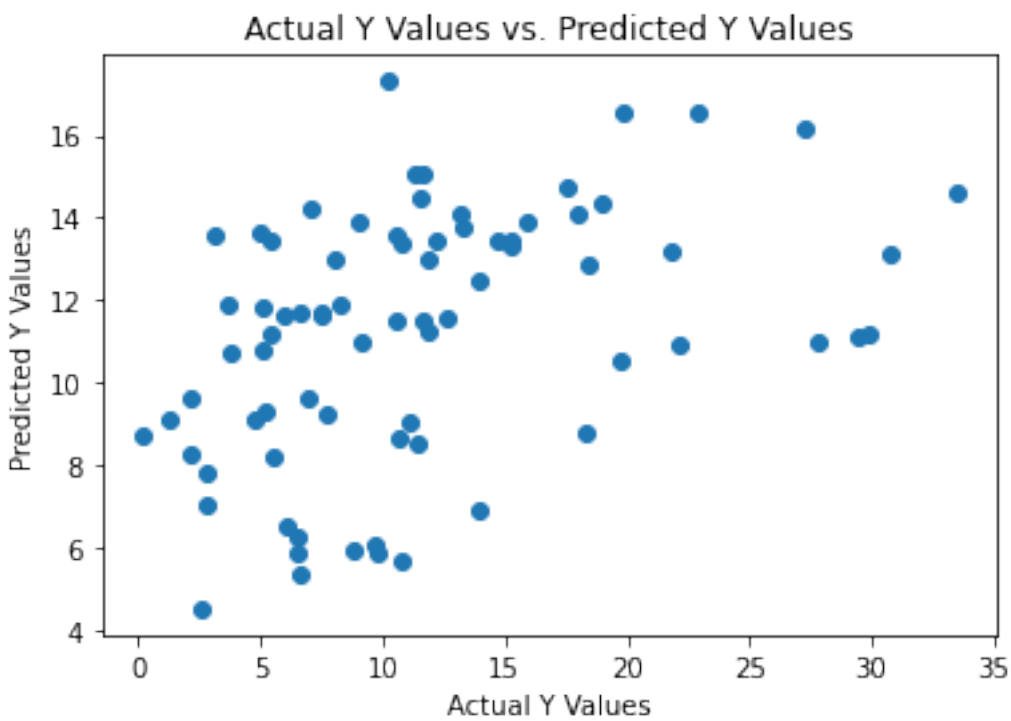
If all other features were held constant, a one unit increase in the Political Regime Score leads to a decrease of 1.170406 in the suicide mortality rate per 100k people.

If all other features were held constant, a one unit increase in the Freedom of the Press Status leads to an increase of the suicide mortality rate per 100k people.

If all other features were held constant, a one unit increase in the Corruption Perception Index leads to a decrease of 9.161802 in the suicide mortality rate per 100k people.

[534]:
```
pred = lm.predict(X_test)
```

[535]:
```
plt.scatter(Y_test, pred)
plt.xlabel('Actual Y Values')
plt.ylabel('Predicted Y Values')
plt.title("Actual Y Values vs. Predicted Y Values")
plt.show()
```



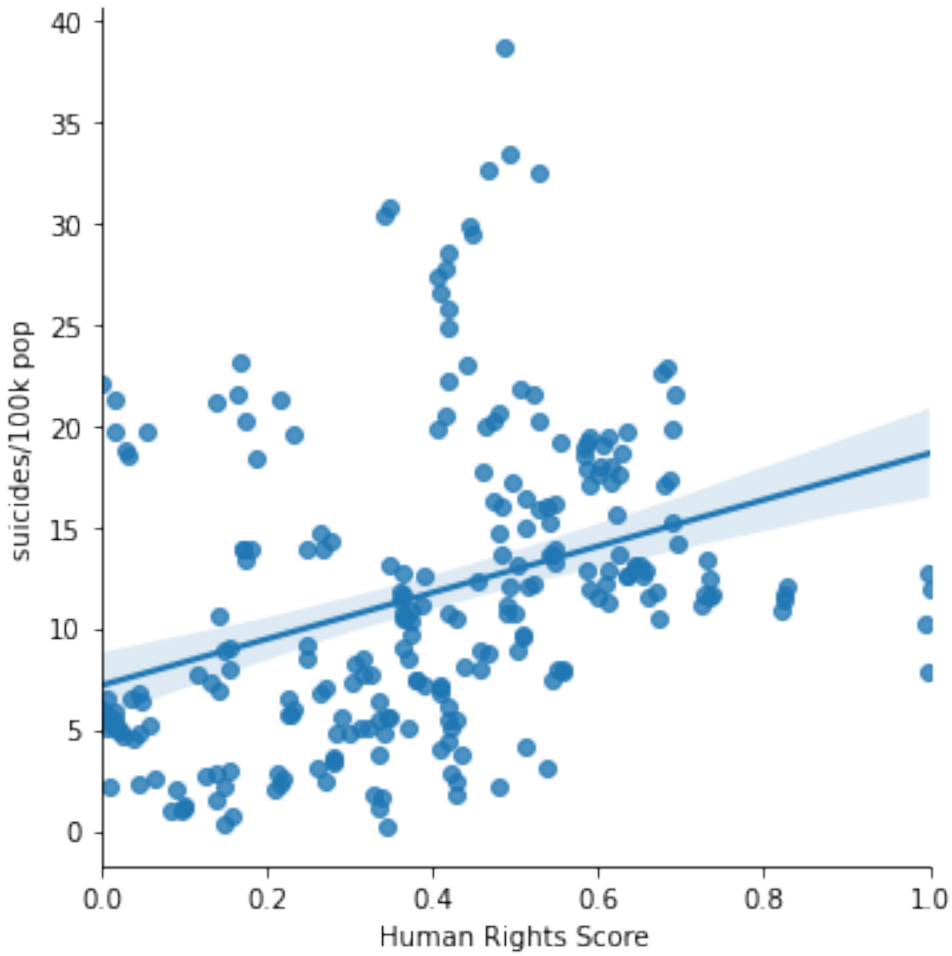
[536]:
```
MSE = metrics.mean_squared_error(Y_test, pred)
print("The Mean Squared Error is", MSE)
```

```
The Mean Squared Error is 46.387282905211066
```

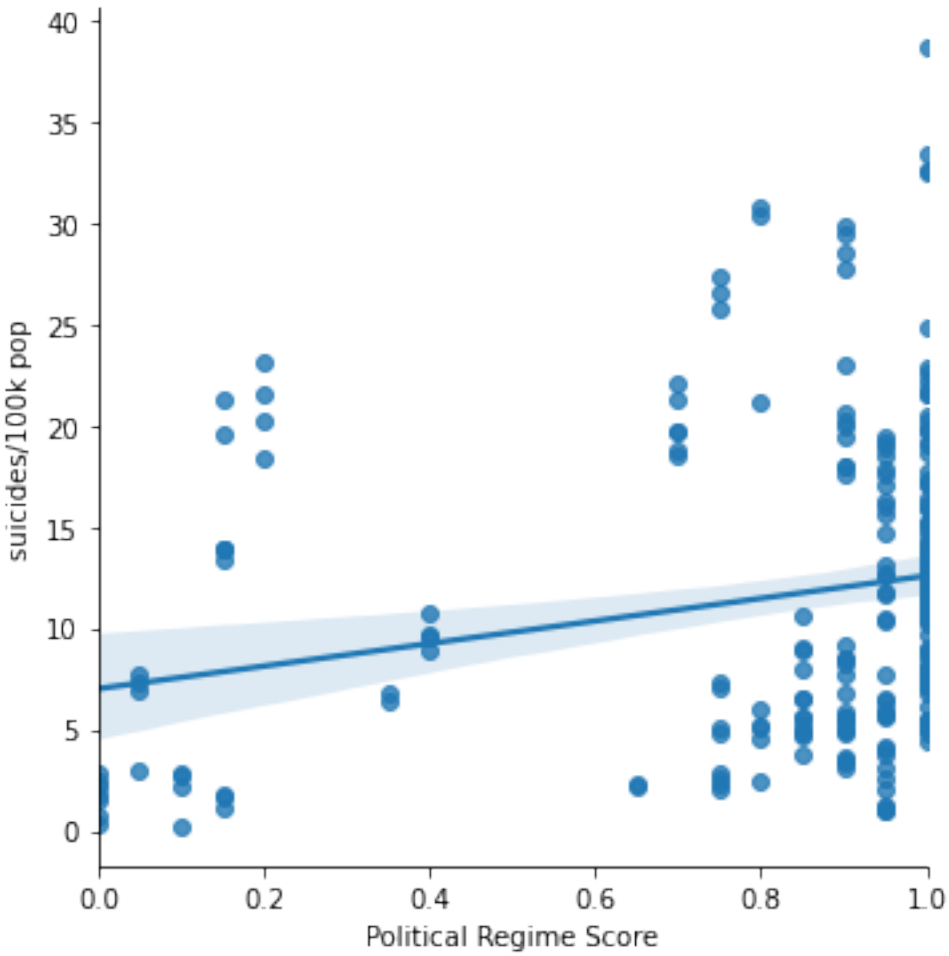
[537]:
```
RMSE = math.sqrt(MSE)
print("The Root Mean Squared Error is", RMSE)
```

```
The Root Mean Squared Error is 6.810821015502541
```

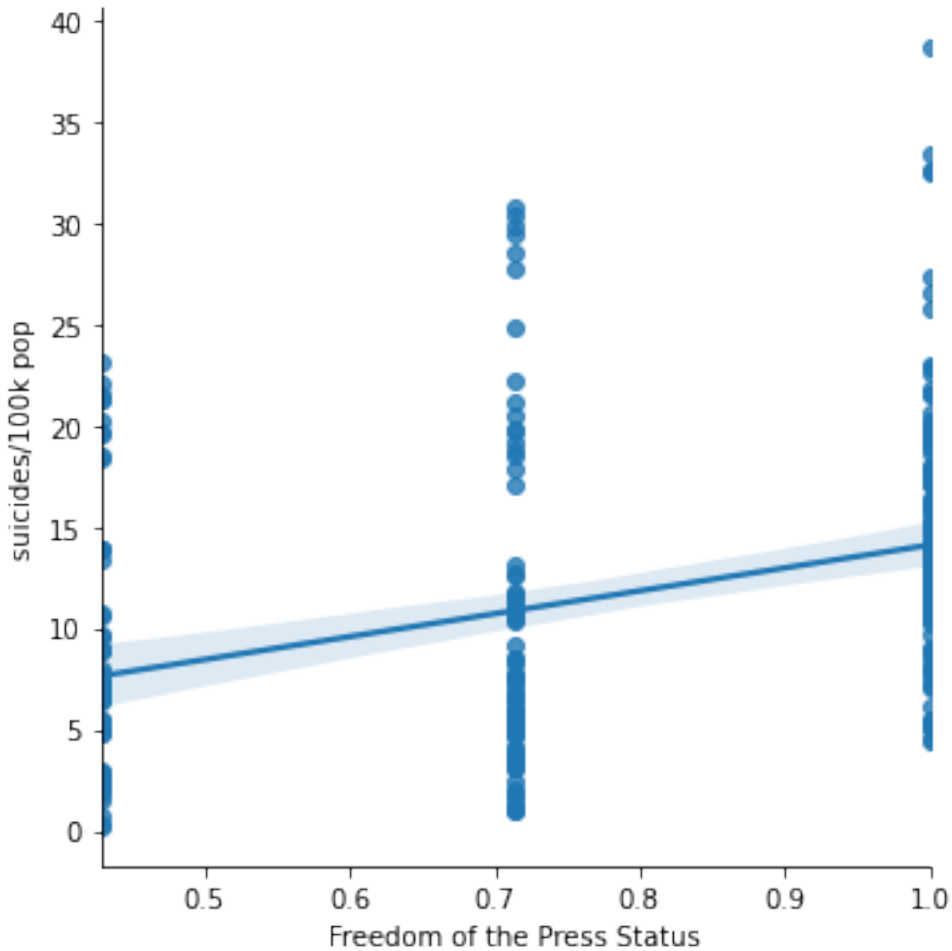
[538]:
```
sns.lmplot(x='Human Rights Score', y='suicides/100k pop', data=normalized_df)
plt.show()
```



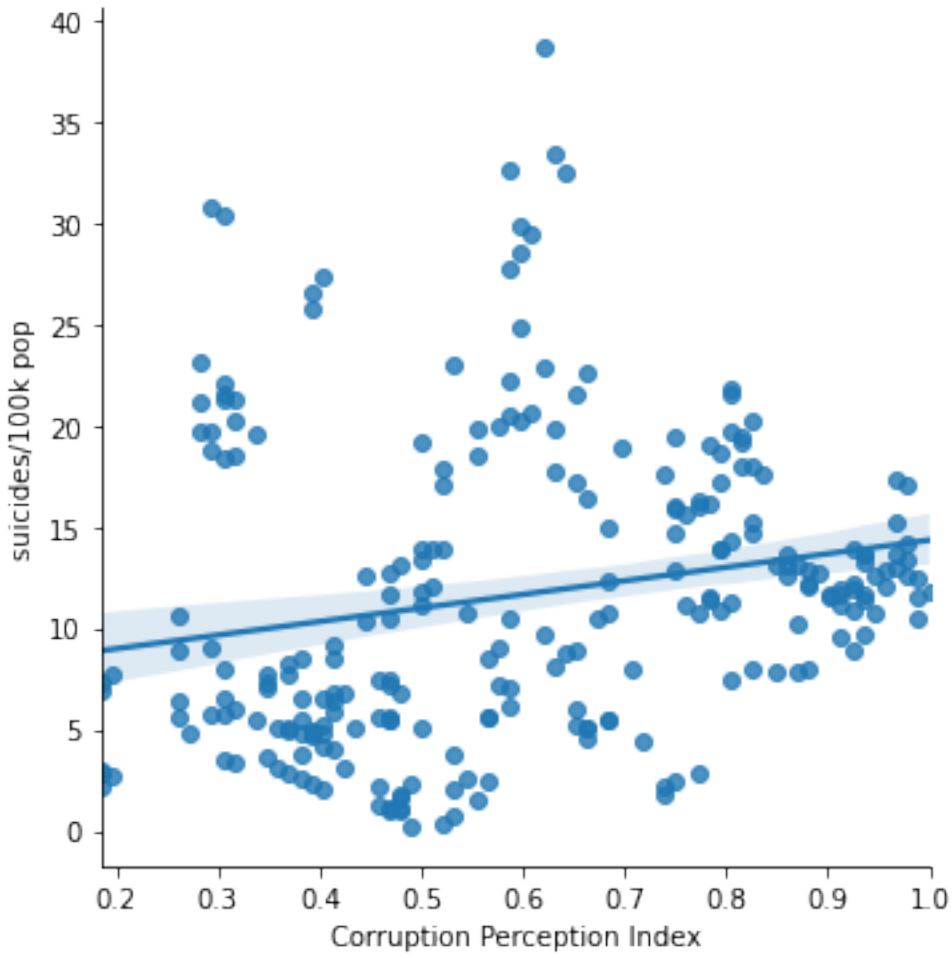
[539]:
```
sns.lmplot(x='Political Regime Score', y='suicides/100k pop',␣
↪data=normalized_df)
plt.show()
```



[540]:
```
sns.lmplot(x='Freedom of the Press Status', y='suicides/100k pop',␣
 ↪data=normalized_df)
plt.show()
```



[541]:
```
sns.lmplot(x='Corruption Perception Index', y='suicides/100k pop',␣
 ↪data=normalized_df)
plt.show()
```



2.3.2 K-Means

[542]:
```
kmeans_data = normalized_df.copy()
```

[543]:
```
kmeans_data.head(2)
```

[543]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.317221                     0.9
1  Argentina  2013            0.314486                     0.9

   Freedom of the Press Status  Corruption Perception Index  suicides/100k pop
0                     0.714286                     0.380435           8.449141
1                     0.714286                     0.369565           7.686740
```

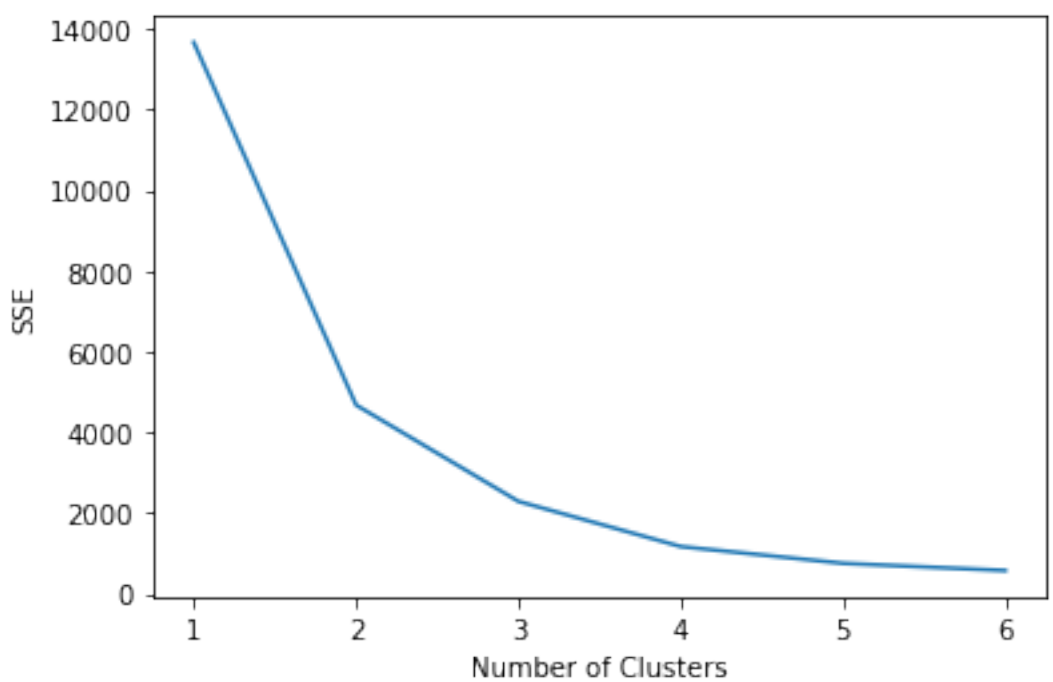
[544]:
```
kmeans_data.drop('Entity', axis=1, inplace=True)
kmeans_data.drop('Year', axis=1, inplace=True)
```

[545]:
```
kmeans_data.head(2)
```

[545]:
```
   Human Rights Score  Political Regime Score  Freedom of the Press Status  \
0            0.317221                     0.9                     0.714286
1            0.314486                     0.9                     0.714286

   Corruption Perception Index  suicides/100k pop
0                     0.380435           8.449141
1                     0.369565           7.686740
```

[546]:
```
numClusters = [1, 2, 3, 4, 5, 6]
SSE = []
for k in numClusters:
    k_means = cluster.KMeans(n_clusters=k)
    k_means.fit(kmeans_data)
    SSE.append(k_means.inertia_)
plt.plot(numClusters, SSE)
plt.xlabel('Number of Clusters')
plt.ylabel('SSE')
```

[546]:
```
Text(0, 0.5, 'SSE')
```



[547]:
```
kmeans = cluster.KMeans(n_clusters=3, max_iter=50, random_state=1)
kmeans.fit(kmeans_data)
```

[547]: KMeans(max_iter=50, n_clusters=3, random_state=1)

[548]:
```
labels = kmeans.labels_
```

[549]:
```
cluster_id = pd.DataFrame(labels, index=normalized_df.Entity, columns=['Cluster␣
↪ID'])
```

[550]:
```
cluster_id.head(15)
```

[550]:
```
            Cluster ID
Entity
Argentina           1
Argentina           1
Argentina           1
Argentina           1
Armenia             1
Armenia             1
Armenia             1
Armenia             1
Australia           2
Australia           2
Australia           2
Australia           2
Austria             2
Austria             2
Austria             2
```

[551]:
```
cluster_id.value_counts()
```

[551]:
```
Cluster ID
1             110
2              92
0              50
dtype: int64
```

[552]:
```
cluster_id.reset_index(drop=True, inplace=True)
```

[553]:
```
cluster_id.head(2)
```

[553]:
```
   Cluster ID
0           1
1           1
```

[554]:
```
normalized_df['Cluster ID'] = cluster_id.values
```

[555]:
```
normalized_df.head(5)
```

[555]:

```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.317221                    0.90
1  Argentina  2013            0.314486                    0.90
2  Argentina  2014            0.306040                    0.90
3  Argentina  2015            0.326683                    0.95
4    Armenia  2012            0.212486                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
0                     0.714286                     0.380435
1                     0.714286                     0.369565
2                     0.714286                     0.369565
3                     0.714286                     0.347826
4                     0.428571                     0.369565

   suicides/100k pop  Cluster ID
0           8.449141           1
1           7.686740           1
2           8.226282           1
3           7.740627           1
4           2.807092           1
```

[556]:

```
Cluster_0 = normalized_df[normalized_df['Cluster ID'] == 0]
Cluster_1 = normalized_df[normalized_df['Cluster ID'] == 1]
Cluster_2 = normalized_df[normalized_df['Cluster ID'] == 2]
```

[557]:

```
Cluster_0.head(2)
```

[557]:

```
     Entity  Year  Human Rights Score  Political Regime Score  \
19  Belarus  2013            0.217333                    0.15
20  Belarus  2014            0.231812                    0.15

    Freedom of the Press Status  Corruption Perception Index  \
19                     0.428571                     0.315217
20                     0.428571                     0.336957

    suicides/100k pop  Cluster ID
19          21.346226           0
20          19.546475           0
```

[558]:

```
Cluster_1.head(2)
```

[558]:

```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.317221                     0.9
1  Argentina  2013            0.314486                     0.9

   Freedom of the Press Status  Corruption Perception Index  \
0                     0.714286                     0.380435
```

```
1                     0.714286                     0.369565

   suicides/100k pop  Cluster ID
0           8.449141           1
1           7.686740           1
```

[559]:
```
Cluster_2.head(2)
```

[559]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
8  Australia  2012            0.520812                     1.0
9  Australia  2013            0.515851                     1.0

   Freedom of the Press Status  Corruption Perception Index  \
8                          1.0                     0.923913
9                          1.0                     0.880435

   suicides/100k pop  Cluster ID
8          12.147517           2
9          12.073973           2
```

[560]:
```
print("The average HRS for Cluster 0 was", round(Cluster_0['Human Rights␣
↪Score'].mean(), 2))
print("The average PRS for Cluster 0 was", round(Cluster_0['Political Regime␣
↪Score'].mean(), 2))
print("The average FPS for Cluster 0 was", round(Cluster_0['Freedom of the␣
↪Press Status'].mean(), 2))
print("The average CPI for Cluster 0 was", round(Cluster_0['Corruption␣
↪Perception Index'].mean(), 2))
print("The average Suicides/100k pop for Cluster 0 was",␣
↪round(Cluster_0['suicides/100k pop'].mean(), 2))
```

```
The average HRS for Cluster 0 was 0.4
The average PRS for Cluster 0 was 0.82
The average FPS for Cluster 0 was 0.8
The average CPI for Cluster 0 was 0.53
The average Suicides/100k pop for Cluster 0 was 23.05
```

[561]:
```
print("The average HRS for Cluster 1 was", round(Cluster_1['Human Rights␣
↪Score'].mean(), 2))
print("The average PRS for Cluster 1 was", round(Cluster_1['Political Regime␣
↪Score'].mean(), 2))
print("The average FPS for Cluster 1 was", round(Cluster_1['Freedom of the␣
↪Press Status'].mean(), 2))
print("The average CPI for Cluster 1 was", round(Cluster_1['Corruption␣
↪Perception Index'].mean(), 2))
print("The average Suicides/100k pop for Cluster 1 was",␣
↪round(Cluster_1['suicides/100k pop'].mean(), 2))
```

```
The average HRS for Cluster 1 was 0.26
The average PRS for Cluster 1 was 0.75
The average FPS for Cluster 1 was 0.67
The average CPI for Cluster 1 was 0.47
The average Suicides/100k pop for Cluster 1 was 5.15
```

[562]:
```
print("The average HRS for Cluster 2 was", round(Cluster_2['Human Rights␣
↪Score'].mean(), 2))
print("The average PRS for Cluster 2 was", round(Cluster_2['Political Regime␣
↪Score'].mean(), 2))
print("The average FPS for Cluster 2 was", round(Cluster_2['Freedom of the␣
↪Press Status'].mean(), 2))
print("The average CPI for Cluster 2 was", round(Cluster_2['Corruption␣
↪Perception Index'].mean(), 2))
print("The average Suicides/100k pop for Cluster 2 was",␣
↪round(Cluster_2['suicides/100k pop'].mean(), 2))
```

```
The average HRS for Cluster 2 was 0.54
The average PRS for Cluster 2 was 0.93
The average FPS for Cluster 2 was 0.91
The average CPI for Cluster 2 was 0.78
The average Suicides/100k pop for Cluster 2 was 13.2
```

Analysis: Cluster 0 had the highest suicide mortality rate at 23.05/100k population. The average Human Rights Score for Cluster 0 was 0.4, which is higher than 0.26 (Cluster 1) but lower than 0.54 (Cluster 2). The average PRS was 0.82, higher than Cluster 1 (0.75) but lower than Cluster 8 (0.93). The average FPS for Cluster 0 was 0.8, higher than Cluster 1 (0.67) but lower than Cluster 2 (0.91). The average CPI for Cluster 0 was 0.53, higher than Cluster 1 (0.47) and lower than Cluster 2 (0.78).

[563]:
```
twelve_df = normalized_df[normalized_df['Year'] == 2012]
thirteen_df = normalized_df[normalized_df['Year'] == 2013]
fourteen_df = normalized_df[normalized_df['Year'] == 2014]
fifteen_df = normalized_df[normalized_df['Year'] == 2015]
```

[564]:
```
twelve_df.head(2)
```

[564]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.317221                    0.90
4    Armenia  2012            0.212486                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
0                     0.714286                     0.380435
4                     0.428571                     0.369565

   suicides/100k pop  Cluster ID
0           8.449141           1
4           2.807092           1
```

[565]:
```
thirteen_df.head(2)
```

[565]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
1  Argentina  2013            0.314486                    0.90
5    Armenia  2013            0.215755                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
1                     0.714286                     0.369565
5                     0.428571                     0.391304

   suicides/100k pop  Cluster ID
1           7.686740           1
5           2.383778           1
```

[566]:
```
fourteen_df.head(2)
```

[566]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
2  Argentina  2014            0.306040                    0.90
6    Armenia  2014            0.210048                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
2                     0.714286                     0.369565
6                     0.428571                     0.402174

   suicides/100k pop  Cluster ID
2           8.226282           1
6           2.069190           1
```

[567]:
```
fifteen_df.head(2)
```

[567]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
3  Argentina  2015            0.326683                    0.95
7    Armenia  2015            0.218647                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
3                     0.714286                     0.347826
7                     0.428571                     0.380435

   suicides/100k pop  Cluster ID
3           7.740627           1
7           2.647268           1
```

[568]:
```
twelve_df.head(35)
```

[568]:
```
          Entity  Year  Human Rights Score  Political Regime Score  \
0      Argentina  2012            0.317221                    0.90
4        Armenia  2012            0.212486                    0.75
8      Australia  2012            0.520812                    1.00
```

12	Austria	2012	0.529400	1.00
16	Bahrain	2012	0.139429	0.00
21	Belgium	2012	0.587734	0.90
25	Brazil	2012	0.016013	0.90
29	Bulgaria	2012	0.388100	0.95
32	Canada	2012	0.589149	1.00
34	Chile	2012	0.359525	1.00
38	Colombia	2012	0.023011	0.85
42	Costa Rica	2012	0.409621	1.00
45	Croatia	2012	0.587243	0.95
49	Cuba	2012	0.172358	0.15
53	Cyprus	2012	0.417958	1.00
57	Denmark	2012	0.652192	1.00
61	Ecuador	2012	0.271242	0.75
65	El Salvador	2012	0.247347	0.90
68	Estonia	2012	0.583741	0.95
72	Finland	2012	0.679007	1.00
76	France	2012	0.474321	0.95
79	Georgia	2012	0.270077	0.80
83	Germany	2012	0.635772	1.00
87	Greece	2012	0.342896	1.00
91	Guatemala	2012	0.259450	0.90
95	Guyana	2012	0.342232	0.80
97	Hungary	2012	0.419268	1.00
101	Ireland	2012	0.586345	1.00
104	Israel	2012	0.000125	0.80
108	Italy	2012	0.378478	1.00
112	Japan	2012	0.506415	1.00
116	Kazakhstan	2012	0.163133	0.20
120	Kuwait	2012	0.337835	0.15
123	Kyrgyzstan	2012	0.142389	0.85
127	Latvia	2012	0.439600	0.90

	Freedom of the Press Status	Corruption Perception Index \
0	0.714286	0.380435
4	0.428571	0.369565
8	1.000000	0.923913
12	1.000000	0.750000
16	0.428571	0.554348
21	1.000000	0.815217
25	0.714286	0.467391
29	0.714286	0.445652
32	1.000000	0.913043
34	0.714286	0.782609
38	0.714286	0.391304
42	1.000000	0.586957
45	0.714286	0.500000

49	0.428571	0.521739
53	1.000000	0.717391
57	1.000000	0.978261
61	0.428571	0.347826
65	0.714286	0.413043
68	1.000000	0.695652
72	1.000000	0.978261
76	1.000000	0.771739
79	0.714286	0.565217
83	1.000000	0.858696
87	0.714286	0.391304
91	0.714286	0.358696
95	0.714286	0.304348
97	0.714286	0.597826
101	1.000000	0.750000
104	0.714286	0.652174
108	0.714286	0.456522
112	1.000000	0.804348
116	0.428571	0.304348
120	0.714286	0.478261
123	0.428571	0.260870
127	1.000000	0.532609

	suicides/100k pop	Cluster ID
0	8.449141	1
4	2.807092	1
8	12.147517	2
12	15.887886	2
16	1.580991	1
21	19.401230	0
25	5.547138	1
29	12.529362	2
32	11.910976	2
34	11.452390	2
38	4.921513	1
42	7.039282	1
45	19.140191	0
49	13.880462	2
53	4.420948	1
57	12.541412	2
61	7.064344	1
65	9.112229	1
68	18.935293	0
72	17.082644	2
76	16.336774	2
79	2.475660	1
83	12.605066	2

```
87            4.819825           1
91            3.149506           1
95           30.322835           0
97           24.848778           0
101          12.841149           2
104           6.060299           1
108           7.495367           1
112          21.814718           0
116          21.566143           0
120           1.601511           1
123          10.623718           2
127          22.987066           0
```

[569]:
```
twelve_df.tail(32)
```

[569]:
```
             Entity  Year  Human Rights Score  Political Regime Score  \
131       Lithuania  2012            0.468208                    1.00
135      Luxembourg  2012            0.995836                    1.00
139       Mauritius  2012            0.374346                    1.00
143          Mexico  2012            0.020145                    0.90
147     Netherlands  2012            0.724318                    1.00
151     New Zealand  2012            0.732343                    1.00
153       Nicaragua  2012            0.232592                    0.95
157          Norway  2012            0.822571                    1.00
162          Panama  2012            0.409519                    0.95
166        Paraguay  2012            0.283248                    0.90
169          Poland  2012            0.461482                    1.00
173        Portugal  2012            0.417154                    1.00
176           Qatar  2012            0.481208                    0.00
180         Romania  2012            0.347041                    0.95
184          Russia  2012            0.000000                    0.70
188       Singapore  2012            0.498863                    0.40
192        Slovakia  2012            0.490902                    1.00
195        Slovenia  2012            0.675891                    1.00
199    South Africa  2012            0.095638                    0.95
203     South Korea  2012            0.447901                    0.90
207           Spain  2012            0.458362                    1.00
211        Suriname  2012            0.407229                    0.75
214          Sweden  2012            0.612591                    1.00
218     Switzerland  2012            0.542734                    1.00
222        Thailand  2012            0.007473                    0.85
226          Turkey  2012            0.089167                    0.95
230    Turkmenistan  2012            0.154703                    0.05
234         Ukraine  2012            0.138312                    0.80
237  United Kingdom  2012            0.542743                    1.00
241   United States  2012            0.246671                    1.00
245         Uruguay  2012            0.604368                    1.00
```

249	Uzbekistan	2012	0.141027	0.05

	Freedom of the Press Status	Corruption Perception Index	\
131	1.000000	0.586957	
135	1.000000	0.869565	
139	1.000000	0.619565	
143	0.428571	0.369565	
147	1.000000	0.913043	
151	1.000000	0.978261	
153	0.714286	0.315217	
157	1.000000	0.923913	
162	0.714286	0.413043	
166	0.428571	0.271739	
169	1.000000	0.630435	
173	1.000000	0.684783	
176	0.428571	0.739130	
180	0.714286	0.478261	
184	0.428571	0.304348	
188	0.428571	0.945652	
192	1.000000	0.500000	
195	1.000000	0.663043	
199	0.714286	0.467391	
203	0.714286	0.608696	
207	1.000000	0.706522	
211	1.000000	0.402174	
214	1.000000	0.956522	
218	1.000000	0.934783	
222	0.428571	0.402174	
226	0.714286	0.532609	
230	0.428571	0.184783	
234	0.714286	0.282609	
237	1.000000	0.804348	
241	1.000000	0.793478	
245	1.000000	0.782609	
249	0.428571	0.184783	

	suicides/100k pop	Cluster ID
131	32.666495	0
135	10.178747	2
139	9.741247	2
143	4.920052	1
147	11.086206	2
151	13.322429	2
153	6.017394	1
157	10.942116	2
162	3.976150	1
166	4.828557	1

```
169          17.682285           2
173          10.728463           2
176           2.146620           1
180          13.102895           2
184          22.011825           0
188          10.794140           2
192          11.159052           2
195          22.610622           0
199           1.025603           1
203          29.485026           0
207           7.985314           1
211          27.312081           0
214          12.895427           2
218          13.665823           2
222           6.599621           1
226           2.102167           1
230           2.942394           1
234          21.106379           0
237           7.442810           1
241          13.863470           2
245          19.004175           0
249           6.955545           1
```

[570]:
```
codes = pd.read_excel('CountryCodes.xlsx')
```

[571]:
```
codes.head(5)
```

[571]:
```
  Code
0  ARG
1  ARM
2  AUS
3  AUT
4  BHR
```

[572]:
```
twelve_df['Code'] = codes.values
```

[573]:
```
twelve_df.head(2)
```

[573]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
0  Argentina  2012            0.317221                    0.90
4    Armenia  2012            0.212486                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
0                     0.714286                     0.380435
4                     0.428571                     0.369565

   suicides/100k pop  Cluster ID Code
```

```
0           8.449141           1  ARG
4           2.807092           1  ARM
```

[574]:
```
init_notebook_mode(connected=True)
```

[575]:
```
data = dict(type='choropleth', locations=twelve_df['Code'], z=␣
↪twelve_df['Cluster ID'],
            text = twelve_df[['Entity','Human Rights Score', 'Political Regime␣
↪Score',
                              'Freedom of the Press Status', 'Corruption␣
↪Perception Index']],
            colorbar = {'title': 'Cluster Identifiers'})
```

[576]:
```
layout = dict(title = 'Countries by Cluster ID (2012)', geo =␣
↪dict(showframe=False, projection={'type':'mercator'}))
```

[577]:
```
choromap3 = go.Figure(data=[data], layout=layout)
```

[578]:
```
iplot(choromap3)
```

[579]:
```
thirteen_df.head(35)
```

[579]:
```
          Entity  Year  Human Rights Score  Political Regime Score  \
1      Argentina  2013            0.314486                    0.90
5        Armenia  2013            0.215755                    0.75
9      Australia  2013            0.515851                    1.00
13       Austria  2013            0.538920                    1.00
17       Bahrain  2013            0.149407                    0.00
19       Belarus  2013            0.217333                    0.15
22       Belgium  2013            0.602544                    0.90
26        Brazil  2013            0.009764                    0.90
30      Bulgaria  2013            0.373656                    0.95
33        Canada  2013            0.607334                    1.00
35         Chile  2013            0.363489                    1.00
39      Colombia  2013            0.027326                    0.85
43    Costa Rica  2013            0.408824                    1.00
46       Croatia  2013            0.587643                    0.95
50          Cuba  2013            0.171338                    0.15
54        Cyprus  2013            0.426893                    1.00
58       Denmark  2013            0.660158                    1.00
62       Ecuador  2013            0.300244                    0.75
66   El Salvador  2013            0.247721                    0.90
69       Estonia  2013            0.597653                    0.95
73       Finland  2013            0.685529                    1.00
77        France  2013            0.484508                    0.95
80       Georgia  2013            0.333616                    0.85
84       Germany  2013            0.644550                    1.00
```

88	Greece	2013	0.368907	1.00
92	Guatemala	2013	0.280542	0.90
96	Guyana	2013	0.347638	0.80
98	Hungary	2013	0.417560	1.00
102	Ireland	2013	0.600464	1.00
105	Israel	2013	0.006085	0.80
109	Italy	2013	0.378504	1.00
113	Japan	2013	0.520428	1.00
117	Kazakhstan	2013	0.166441	0.20
121	Kuwait	2013	0.333397	0.15
124	Kyrgyzstan	2013	0.148295	0.85

	Freedom of the Press Status	Corruption Perception Index \
1	0.714286	0.369565
5	0.428571	0.391304
9	1.000000	0.880435
13	1.000000	0.750000
17	0.428571	0.521739
19	0.428571	0.315217
22	1.000000	0.815217
26	0.714286	0.456522
30	0.714286	0.445652
33	1.000000	0.880435
35	0.714286	0.771739
39	0.714286	0.391304
43	1.000000	0.576087
46	0.714286	0.521739
50	0.428571	0.500000
54	1.000000	0.684783
58	1.000000	0.989130
62	0.428571	0.380435
66	0.714286	0.413043
69	1.000000	0.739130
73	1.000000	0.967391
77	1.000000	0.771739
80	0.714286	0.532609
84	1.000000	0.847826
88	0.714286	0.434783
92	0.714286	0.315217
96	0.714286	0.293478
98	0.714286	0.586957
102	1.000000	0.782609
105	1.000000	0.663043
109	0.714286	0.467391
113	1.000000	0.804348
117	0.428571	0.282609
121	0.714286	0.467391

```
124                     0.428571                     0.260870

     suicides/100k pop  Cluster ID
1             7.686740           1
5             2.383778           1
9            12.073973           2
13           15.978484           2
17            0.411695           1
19           21.346226           0
22           18.002785           2
26            5.608164           1
30           10.418106           2
33           12.168753           2
35           10.700492           2
39            4.748969           1
43            7.153982           1
46           17.109837           2
50           13.889030           2
54            5.534946           1
58           11.510090           2
62            4.809809           1
66            8.555394           1
69           17.536860           2
73           17.272156           2
77           16.024018           2
80            3.744693           1
84           13.059235           2
88            5.109627           1
92            3.335318           1
96           30.707725           0
98           22.175487           0
102          11.520143           2
105           5.148071           1
109           7.460437           1
113          21.566789           0
117          23.081905           0
121           1.147145           1
124           8.898288           1
```

[580]:
```
thirteen_df.tail(32)
```

[580]:
```
             Entity  Year  Human Rights Score  Political Regime Score  \
128          Latvia  2013            0.462912                    0.90
132       Lithuania  2013            0.485528                    1.00
136      Luxembourg  2013            0.998852                    1.00
140       Mauritius  2013            0.370488                    1.00
144          Mexico  2013            0.017931                    0.90
```

148	Netherlands	2013	0.728913	1.00
152	New Zealand	2013	0.735650	1.00
154	Nicaragua	2013	0.225732	0.95
158	Norway	2013	0.822664	1.00
163	Panama	2013	0.433667	0.95
167	Paraguay	2013	0.290584	0.95
170	Poland	2013	0.495313	1.00
174	Portugal	2013	0.427534	1.00
177	Qatar	2013	0.429506	0.00
181	Romania	2013	0.364755	0.95
185	Russia	2013	0.016523	0.70
189	Singapore	2013	0.507932	0.40
193	Slovakia	2013	0.493783	1.00
196	Slovenia	2013	0.682401	1.00
200	South Africa	2013	0.100414	0.95
204	South Korea	2013	0.444370	0.90
208	Spain	2013	0.466772	1.00
212	Suriname	2013	0.409264	0.75
215	Sweden	2013	0.623809	1.00
219	Switzerland	2013	0.547307	1.00
223	Thailand	2013	0.035633	0.85
227	Turkey	2013	0.063804	0.95
231	Turkmenistan	2013	0.147863	0.10
238	United Kingdom	2013	0.552219	1.00
242	United States	2013	0.267045	1.00
246	Uruguay	2013	0.616202	1.00
250	Uzbekistan	2013	0.132285	0.05

	Freedom of the Press Status	Corruption Perception Index	\
128	1.000000	0.576087	
132	1.000000	0.619565	
136	1.000000	0.869565	
140	1.000000	0.565217	
144	0.428571	0.369565	
148	1.000000	0.902174	
152	1.000000	0.989130	
154	0.714286	0.304348	
158	1.000000	0.934783	
163	0.714286	0.380435	
167	0.714286	0.260870	
170	1.000000	0.652174	
174	1.000000	0.673913	
177	0.428571	0.739130	
181	0.714286	0.467391	
185	0.428571	0.304348	
189	0.428571	0.934783	
193	1.000000	0.510870	

196	1.000000	0.619565
200	0.714286	0.456522
204	0.714286	0.597826
208	1.000000	0.641304
212	1.000000	0.391304
215	1.000000	0.967391
219	1.000000	0.923913
223	0.428571	0.380435
227	0.428571	0.543478
231	0.428571	0.184783
238	1.000000	0.826087
242	1.000000	0.793478
246	1.000000	0.793478
250	0.428571	0.184783

	suicides/100k pop	Cluster ID
128	20.026521	0
132	38.658504	0
136	7.802001	1
140	8.541124	1
144	5.143344	1
148	11.673317	2
152	12.421999	2
154	6.589155	1
158	11.620090	2
163	3.760065	1
167	5.582241	1
170	17.260862	2
174	10.535726	2
177	1.732738	1
181	12.666191	2
185	21.296976	0
189	9.750220	2
193	12.026061	2
196	22.892034	0
200	1.218027	1
204	29.899546	0
208	8.756970	1
212	26.606602	0
215	13.630197	2
219	13.942920	2
223	6.518396	1
227	2.590142	1
231	2.127187	1
238	8.027719	1
242	13.931532	2
246	17.252984	2

```
250           7.265567           1
```

[581]:
```
codes13 = pd.read_excel('CountryCodes13.xlsx')
```

[582]:
```
codes13.head(2)
```

[582]:
```
  Code
0  ARG
1  ARM
```

[583]:
```
thirteen_df['Code'] = codes13.values
```

[584]:
```
thirteen_df.head(2)
```

[584]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
1  Argentina  2013            0.314486                    0.90
5    Armenia  2013            0.215755                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
1                     0.714286                     0.369565
5                     0.428571                     0.391304

   suicides/100k pop  Cluster ID Code
1           7.686740           1  ARG
5           2.383778           1  ARM
```

[585]:
```
data13 = dict(type='choropleth', locations=thirteen_df['Code'], z=␣
↪thirteen_df['Cluster ID'],
            text = thirteen_df[['Entity','Human Rights Score', 'Political␣
↪Regime Score',
                              'Freedom of the Press Status', 'Corruption␣
↪Perception Index']],
            colorbar = {'title': 'Cluster Identifiers'})
```

[586]:
```
layout13 = dict(title = 'Countries by Cluster ID (2013)', geo =␣
↪dict(showframe=False, projection={'type':'mercator'}))
```

[587]:
```
choromap3_13 = go.Figure(data=[data13], layout=layout13)
```

[588]:
```
iplot(choromap3_13)
```

[589]:
```
fourteen_df.head(33)
```

[589]:
```
          Entity  Year  Human Rights Score  Political Regime Score  \
2      Argentina  2014            0.306040                    0.90
6        Armenia  2014            0.210048                    0.75
10     Australia  2014            0.503725                    1.00
14       Austria  2014            0.546179                    1.00
```

18	Bahrain	2014	0.158202	0.00
20	Belarus	2014	0.231812	0.15
23	Belgium	2014	0.615886	0.90
27	Brazil	2014	0.008015	0.90
31	Bulgaria	2014	0.364421	0.95
36	Chile	2014	0.374097	1.00
40	Colombia	2014	0.044468	0.85
44	Costa Rica	2014	0.419414	1.00
47	Croatia	2014	0.587636	0.95
51	Cuba	2014	0.173860	0.15
55	Cyprus	2014	0.418197	1.00
59	Denmark	2014	0.668537	1.00
63	Ecuador	2014	0.311939	0.75
67	El Salvador	2014	0.263568	0.90
70	Estonia	2014	0.613131	0.95
74	Finland	2014	0.690152	1.00
78	France	2014	0.479844	0.95
81	Georgia	2014	0.345561	0.85
85	Germany	2014	0.651260	1.00
89	Greece	2014	0.336546	1.00
93	Guatemala	2014	0.281734	0.90
99	Hungary	2014	0.416443	1.00
103	Ireland	2014	0.611993	1.00
106	Israel	2014	0.014310	0.80
110	Italy	2014	0.390339	1.00
114	Japan	2014	0.526570	1.00
118	Kazakhstan	2014	0.173842	0.20
122	Kuwait	2014	0.327891	0.15
125	Kyrgyzstan	2014	0.153724	0.85

	Freedom of the Press Status	Corruption Perception Index	\
2	0.714286	0.369565	
6	0.428571	0.402174	
10	1.000000	0.869565	
14	1.000000	0.782609	
18	0.428571	0.532609	
20	0.428571	0.336957	
23	1.000000	0.826087	
27	0.714286	0.467391	
31	0.714286	0.467391	
36	0.714286	0.793478	
40	0.714286	0.402174	
44	1.000000	0.586957	
47	0.714286	0.521739	
51	0.428571	0.500000	
55	1.000000	0.684783	
59	1.000000	1.000000	

63	0.428571	0.358696
67	0.714286	0.423913
70	1.000000	0.750000
74	1.000000	0.967391
78	1.000000	0.750000
81	0.714286	0.565217
85	1.000000	0.858696
89	0.714286	0.467391
93	0.714286	0.347826
99	0.714286	0.586957
103	1.000000	0.804348
106	1.000000	0.652174
110	0.714286	0.467391
114	1.000000	0.826087
118	0.428571	0.315217
122	0.714286	0.478261
125	0.428571	0.293478

	suicides/100k pop	Cluster ID
2	8.226282	1
6	2.069190	1
10	13.184305	2
14	16.140196	2
18	0.741372	1
20	19.546475	0
23	17.950915	2
27	5.616382	1
31	10.517334	2
36	10.903971	2
40	4.829425	1
44	6.085827	1
47	17.913624	2
51	13.437018	2
55	5.467075	1
59	11.780866	2
63	5.134610	1
67	6.776294	1
70	19.406499	0
74	15.290864	2
78	14.733902	2
81	5.589812	1
85	13.174324	2
89	5.442873	1
93	3.670990	1
99	20.450901	0
103	11.229094	2
106	5.185812	1

```
110           7.137611           1
114          20.251260           0
118          20.215384           0
122           1.830277           1
125           9.045548           1
```

[590]:
```
fourteen_df.tail(33)
```

[590]:
```
             Entity  Year  Human Rights Score  Political Regime Score  \
129          Latvia  2014            0.472117                    0.90
133       Lithuania  2014            0.493302                    1.00
137      Luxembourg  2014            0.998845                    1.00
141       Mauritius  2014            0.368442                    1.00
145          Mexico  2014            0.001967                    0.90
149     Netherlands  2014            0.732818                    1.00
155       Nicaragua  2014            0.229983                    0.95
159          Norway  2014            0.825187                    1.00
161            Oman  2014            0.343046                    0.10
164          Panama  2014            0.512942                    0.95
168        Paraguay  2014            0.333888                    0.95
171          Poland  2014            0.510712                    1.00
175        Portugal  2014            0.452904                    1.00
178           Qatar  2014            0.428173                    0.00
182         Romania  2014            0.364261                    0.95
186          Russia  2014            0.016741                    0.70
190       Singapore  2014            0.508811                    0.40
194        Slovakia  2014            0.488668                    1.00
197        Slovenia  2014            0.688669                    1.00
201    South Africa  2014            0.101260                    0.95
205     South Korea  2014            0.419821                    0.90
209           Spain  2014            0.455849                    1.00
213        Suriname  2014            0.420060                    0.75
216          Sweden  2014            0.634671                    1.00
220     Switzerland  2014            0.548752                    1.00
224        Thailand  2014            0.048475                    0.35
228          Turkey  2014            0.046574                    0.65
232    Turkmenistan  2014            0.139751                    0.10
235         Ukraine  2014            0.053964                    0.70
239  United Kingdom  2014            0.558300                    1.00
243   United States  2014            0.277163                    1.00
247         Uruguay  2014            0.626640                    1.00
251      Uzbekistan  2014            0.116519                    0.05

     Freedom of the Press Status  Corruption Perception Index  \
129                     1.000000                     0.597826
133                     1.000000                     0.630435
137                     1.000000                     0.891304
```

141	1.000000	0.586957
145	0.428571	0.380435
149	1.000000	0.902174
155	0.714286	0.304348
159	1.000000	0.934783
161	0.428571	0.489130
164	0.714286	0.402174
168	0.714286	0.260870
171	1.000000	0.663043
175	1.000000	0.684783
178	0.428571	0.750000
182	0.714286	0.467391
186	0.428571	0.293478
190	0.428571	0.913043
194	1.000000	0.543478
197	1.000000	0.630435
201	0.714286	0.478261
205	0.714286	0.597826
209	1.000000	0.652174
213	1.000000	0.391304
216	1.000000	0.945652
220	1.000000	0.934783
224	0.428571	0.413043
228	0.428571	0.489130
232	0.428571	0.184783
235	0.714286	0.282609
239	1.000000	0.847826
243	1.000000	0.804348
247	1.000000	0.793478
251	0.428571	0.195652

	suicides/100k pop	Cluster ID
129	20.168518	0
133	33.434271	0
137	12.763363	2
141	10.421878	2
145	5.482171	1
149	11.566396	2
155	5.756675	1
159	11.352388	2
161	0.192779	1
164	4.175884	1
168	6.448217	1
171	16.456076	2
175	12.288282	2
178	2.442618	1
182	11.655617	2

```
186          19.701744           0
190           9.622239           2
194          10.799881           2
197          19.879229           0
201           1.185800           1
205          28.543770           0
209           8.853871           1
213          25.718253           0
216          12.595282           2
220          13.214470           2
224           6.463862           1
228           2.279667           1
232           2.826722           1
235          19.691497           0
239           7.904841           1
243          14.364077           2
247          18.598679           0
251           7.670198           1
```

[591]:
```
codes14 = pd.read_excel('CountryCodes14.xlsx')
```

[592]:
```
codes14.head(5)
```

[592]:
```
  Code
0  ARG
1  ARM
2  AUS
3  AUT
4  BHR
```

[593]:
```
fourteen_df['Code'] = codes14.values
```

[594]:
```
fourteen_df.head(2)
```

[594]:
```
      Entity  Year  Human Rights Score  Political Regime Score  \
2  Argentina  2014            0.306040                    0.90
6    Armenia  2014            0.210048                    0.75

   Freedom of the Press Status  Corruption Perception Index  \
2                     0.714286                     0.369565
6                     0.428571                     0.402174

   suicides/100k pop  Cluster ID Code
2           8.226282           1  ARG
6           2.069190           1  ARM
```

[595]:
```
data14 = dict(type='choropleth', locations=fourteen_df['Code'], z=␣
↪fourteen_df['Cluster ID'],
            text = fourteen_df[['Entity','Human Rights Score', 'Political␣
↪Regime Score',
                                'Freedom of the Press Status', 'Corruption␣
↪Perception Index']],
            colorbar = {'title': 'Cluster Identifiers'})
```

[596]:
```
layout14 = dict(title = 'Countries by Cluster ID (2014)', geo =␣
↪dict(showframe=False, projection={'type':'mercator'}))
```

[597]:
```
choromap3_14 = go.Figure(data=[data14], layout=layout14)
```

[598]:
```
iplot(choromap3_14)
```

2.3.3 Decision Tree Classifier

This Classifier would be based on: Human Rights, Freedom, Corruption... etc as factors

[623]:
```
Data_HRS
```

[623]:
```
         Entity  Year  Human Rights Score  Political Regime Score  \
0     Argentina  2012            0.317221                    0.90
1     Argentina  2013            0.314486                    0.90
2     Argentina  2014            0.306040                    0.90
3     Argentina  2015            0.326683                    0.95
4       Armenia  2012            0.212486                    0.75
..          …   …                  …                       …
247     Uruguay  2014            0.626640                    1.00
248     Uruguay  2015            0.633566                    1.00
249  Uzbekistan  2012            0.141027                    0.05
250  Uzbekistan  2013            0.132285                    0.05
251  Uzbekistan  2014            0.116519                    0.05

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                            0.5                     0.240000         3248
1                            0.5                     0.226667         2987
2                            0.5                     0.226667         3231
3                            0.5                     0.200000         3073
4                            0.0                     0.226667           79
..                           …                        …          …
247                          1.0                     0.746667          591
248                          1.0                     0.760000          630
249                          0.0                     0.000000         1835
250                          0.0                     0.000000         1950
251                          0.0                     0.013333         2095

     suicides/100k pop
```

```
0             8.449141
1             7.686740
2             8.226282
3             7.740627
4             2.807092
..                 …
247          18.598679
248          19.744296
249           6.955545
250           7.265567
251           7.670198

[252 rows x 8 columns]
```

[624]:
```python
cat = ['Entity', 'Year']
l = sklearn.preprocessing.LabelEncoder()
for c in cat:
    Data_HRS[c] = l.fit_transform(Data_HRS[c])

Data_HRS
```

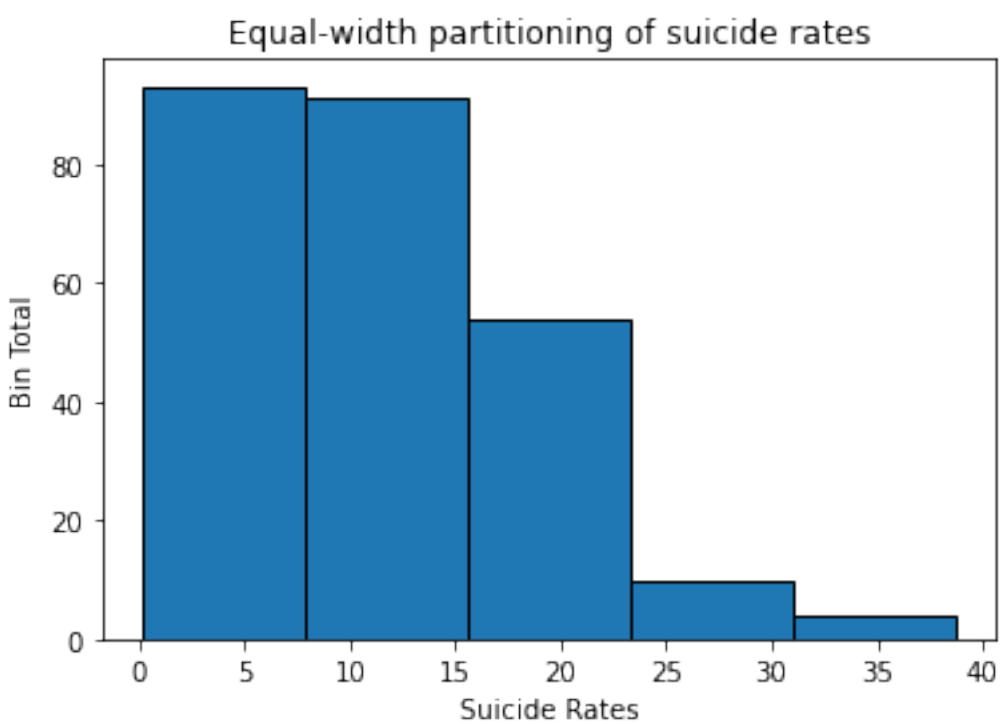
[624]:
```
     Entity  Year  Human Rights Score  Political Regime Score  \
0         0     0            0.317221                    0.90
1         0     1            0.314486                    0.90
2         0     2            0.306040                    0.90
3         0     3            0.326683                    0.95
4         1     0            0.212486                    0.75
..      …   …                  …                       …
247      67     2            0.626640                    1.00
248      67     3            0.633566                    1.00
249      68     0            0.141027                    0.05
250      68     1            0.132285                    0.05
251      68     2            0.116519                    0.05

     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
0                            0.5                     0.240000         3248
1                            0.5                     0.226667         2987
2                            0.5                     0.226667         3231
3                            0.5                     0.200000         3073
4                            0.0                     0.226667           79
..                           …                          …          …
247                          1.0                     0.746667          591
248                          1.0                     0.760000          630
249                          0.0                     0.000000         1835
250                          0.0                     0.000000         1950
251                          0.0                     0.013333         2095
```

```
     suicides/100k pop
0             8.449141
1             7.686740
2             8.226282
3             7.740627
4             2.807092
..                 …
247          18.598679
248          19.744296
249           6.955545
250           7.265567
251           7.670198

[252 rows x 8 columns]
```

Creating 4 classes based on suicide rate: Very low, Low, Medium, and High

[626]:
```
suicides_list = Data_HRS['suicides/100k pop'].tolist()
suicides_list = sorted(suicides_list)
```

[627]:
```
plt.hist(suicides_list, bins = 5, edgecolor = "black")
plt.title("Equal-width partitioning of suicide rates")
plt.xlabel("Suicide Rates")
plt.ylabel("Bin Total")
plt.show()
```



[628]:
```
suicides_list_range = round(max(suicides_list) - min(suicides_list), 2)
print("The range is", suicides_list_range)
length = round((suicides_list_range / 5), 2)
```

```
The range is 38.47
```

[629]:
```
cut_1 = round(min(suicides_list) + length, 2)
print("Cut 1 is", cut_1)
cut_2 = round(cut_1 + length, 2)
print("Cut 2 is", cut_2)
cut_3 = round(cut_2 + length, 2)
print("Cut 3 is", cut_3)
```

```
Cut 1 is 7.88
Cut 2 is 15.57
Cut 3 is 23.26
```

[630]:
```
Class1 = Data_HRS[Data_HRS['suicides/100k pop'] <= 7.88]
Class1['Class'] = "Very Low"
Class2 = Data_HRS[(Data_HRS['suicides/100k pop'] > 7.88) & (Data_HRS['suicides/
↪100k pop'] <= 15.57)]
Class2['Class'] = "Low"
Class3 = Data_HRS[(Data_HRS['suicides/100k pop'] > 15.57) & (Data_HRS['suicides/
↪100k pop'] <= 23.26)]
Class3['Class'] = 'Medium'
Class4 = Data_HRS[Data_HRS['suicides/100k pop'] > 23.26]
Class4['Class'] = 'High'
```

[631]:
```
Concat1 = pd.concat([Class1, Class2], axis=0)
Concat2 = pd.concat([Concat1, Class3], axis=0)
suicide_tree_HRS = pd.concat([Concat2, Class4], axis=0)
```

[632]:
```
suicide_tree_HRS
```

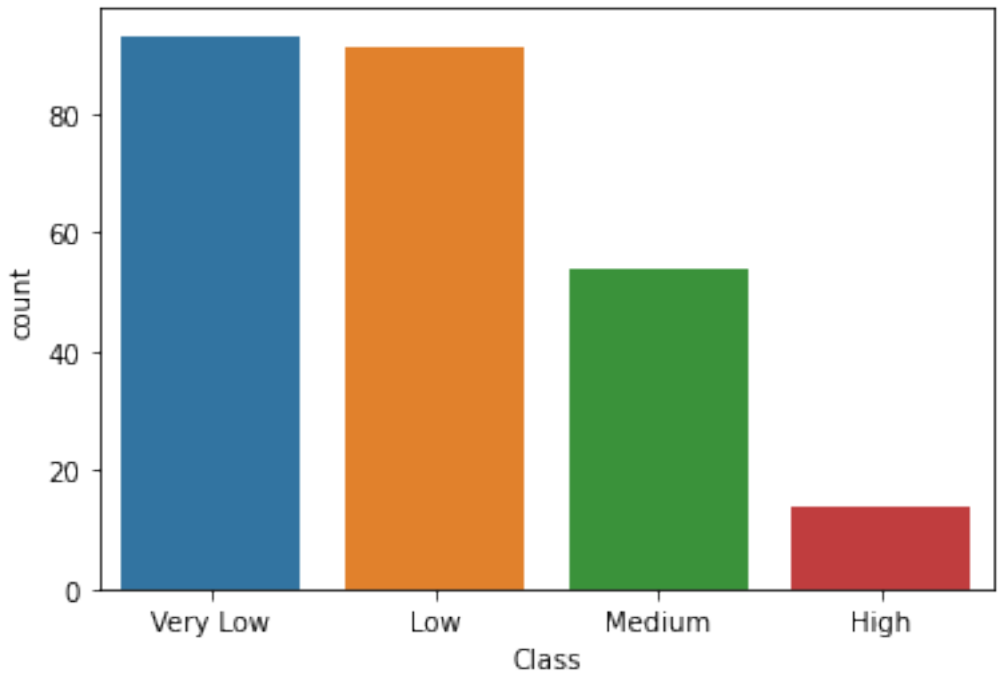
[632]:
```
     Entity  Year  Human Rights Score  Political Regime Score  \
1         0     1            0.314486                    0.90
3         0     3            0.326683                    0.95
4         1     0            0.212486                    0.75
5         1     1            0.215755                    0.75
6         1     2            0.210048                    0.75
..      …   …                 …                       …
205      56     2            0.419821                    0.90
206      56     3            0.414270                    0.90
211      58     0            0.407229                    0.75
212      58     1            0.409264                    0.75
213      58     2            0.420060                    0.75
```

```
     Freedom of the Press Status  Corruption Perception Index  suicides_no  \
1                            0.5                     0.226667         2987
3                            0.5                     0.200000         3073
4                            0.0                     0.226667           79
5                            0.0                     0.253333           67
6                            0.0                     0.266667           58
..                           …                          …          …
205                          0.5                     0.506667        13834
206                          0.5                     0.493333        13510
211                          1.0                     0.266667          133
212                          1.0                     0.253333          131
213                          1.0                     0.253333          128

     suicides/100k pop     Class
1             7.686740  Very Low
3             7.740627  Very Low
4             2.807092  Very Low
5             2.383778  Very Low
6             2.069190  Very Low
..                 …       …
205          28.543770      High
206          27.757374      High
211          27.312081      High
212          26.606602      High
213          25.718253      High

[252 rows x 9 columns]
```

Checking and fixing imbalanced classes

[633]:
```
ax = sns.countplot(x=suicide_tree_HRS['Class'], data=suicide_tree_HRS)
print(suicide_tree_HRS['Class'].value_counts())
```

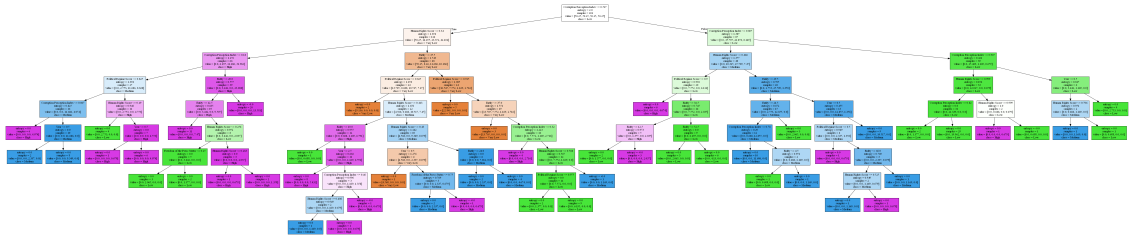
```
Very Low    93
Low         91
Medium      54
High        14
Name: Class, dtype: int64
```



DecisionTree Classifier

[640]:
```
y = suicide_tree_HRS['Class']
x = suicide_tree_HRS.drop(['Class', 'suicides/100k pop', 'suicides_no'], axis=1)

X_train, X_test, Y_train, Y_test = train_test_split(x, y, test_size=0.2,
                                                    random_state=1,
                                                    ␣
 ↪stratify=suicide_tree_HRS['Class'])
clf_train = tree.DecisionTreeClassifier(criterion='entropy',␣
 ↪class_weight='balanced')
clf_train = clf_train.fit(X_train, Y_train)

dot_data_train = tree.export_graphviz(clf_train, feature_names=x.columns,
                                      class_names=['Very Low', 'Low','Medium',␣
 ↪'High'],
                                      filled=True, out_file=None)
graph_train = pydotplus.graph_from_dot_data(dot_data_train)
Image(graph_train.create_png())
```

[640]:

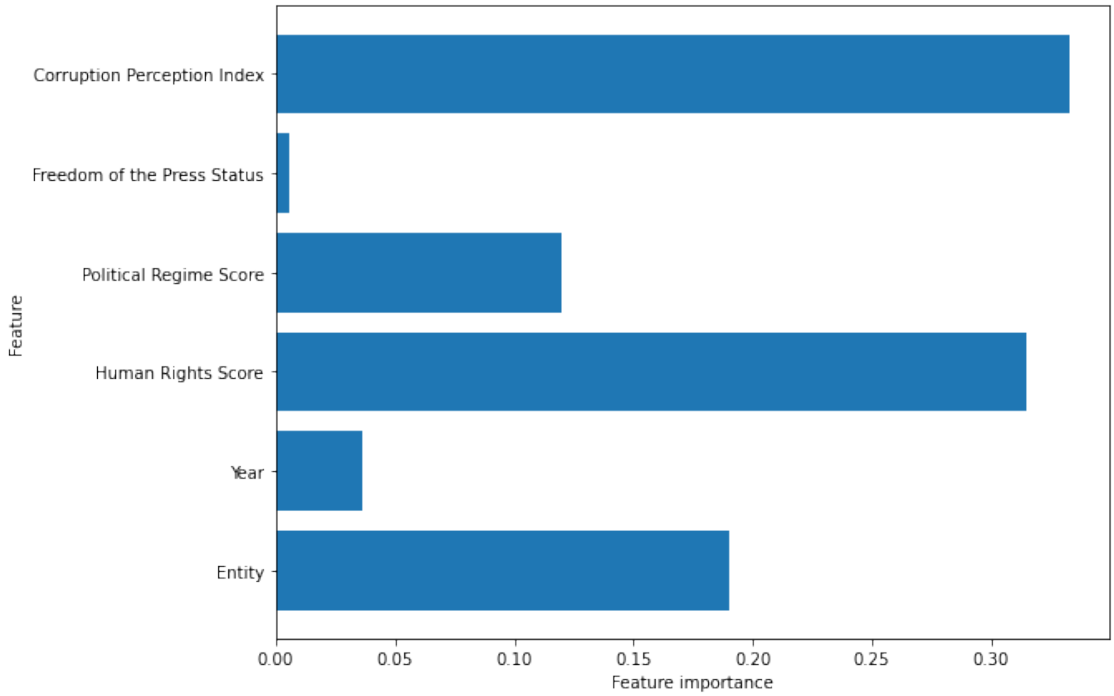


Checking accuracy and the features importance

[641]:
```
Y_predTrain = clf_train.predict(X_train)
Y_predTest = clf_train.predict(X_test)
trainAcc = accuracy_score(Y_train, Y_predTrain)
testAcc = accuracy_score(Y_test, Y_predTest)
print('Accuracy on train data is %.2f' % trainAcc)
print('Accuracy on test data is %.2f' % testAcc)
```

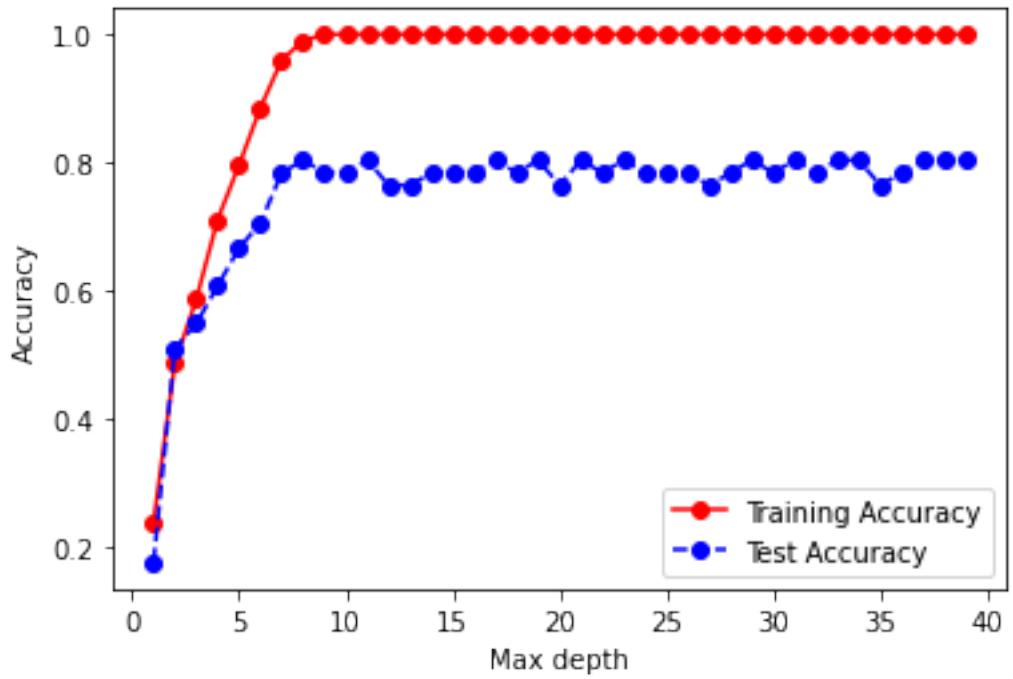
```
Accuracy on train data is 1.00
Accuracy on test data is 0.78
```

[642]:
```
plt.figure(figsize=(9,7))
n_features = X_train.shape[1]
plt.barh(range(n_features), clf_train.feature_importances_, align='center')
plt.yticks(np.arange(n_features), X_train.columns)
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.show()
```



[643]:
```
L=[i for i in np.arange(1,160,5)]
maxdepths = [i for i in np.arange(1,40,1)]
trainAcc = np.zeros(len(maxdepths))
testAcc = np.zeros(len(maxdepths))
index = 0
for depth in maxdepths:
    clf = tree.DecisionTreeClassifier(class_weight='balanced', max_depth=depth)
    clf = clf.fit(X_train, Y_train)
    trainAcc[index] = accuracy_score(Y_train, clf.predict(X_train))
    testAcc[index] = accuracy_score(Y_test, clf.predict(X_test))
    index += 1
plt.plot(maxdepths,trainAcc,'ro-',maxdepths,testAcc,'bo--')
plt.legend(['Training Accuracy','Test Accuracy'])
plt.xlabel('Max depth')
plt.ylabel('Accuracy')
```

[643]: Text(0, 0.5, 'Accuracy')



2.3.4 An Additional Classifier (just for more practice): Based on GDP, Gender, Age group as factors

Some Preprocess for the data before making our model

[599]:
```
df_suicide_tree = pd.read_csv('master.csv')
```

[600]:
```
df_suicide_tree.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27820 entries, 0 to 27819
Data columns (total 12 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   country             27820 non-null  object
 1   year                27820 non-null  int64
 2   sex                 27820 non-null  object
 3   age                 27820 non-null  object
 4   suicides_no         27820 non-null  int64
 5   population          27820 non-null  int64
 6   suicides/100k pop   27820 non-null  float64
 7   country-year        27820 non-null  object
 8   HDI for year        8364 non-null   float64
 9    gdp_for_year ($)   27820 non-null  object
 10  gdp_per_capita ($)  27820 non-null  int64
 11  generation          27820 non-null  object
```

```
dtypes: float64(2), int64(4), object(6)
memory usage: 2.5+ MB
```

[601]:
```
df_suicide_tree.isnull().sum()
```

[601]:
```
country                   0
year                      0
sex                       0
age                       0
suicides_no               0
population                0
suicides/100k pop         0
country-year              0
HDI for year          19456
 gdp_for_year ($)         0
gdp_per_capita ($)        0
generation                0
dtype: int64
```

[602]:
```
df_suicide_tree = df_suicide_tree.drop(['HDI for year'], axis = 1)
df_suicide_tree = df_suicide_tree.drop(['country-year'], axis = 1)
df_suicide_tree = df_suicide_tree.dropna()
```

[603]:
```
cat = ['country', 'year','age', 'sex', 'generation']
l = sklearn.preprocessing.LabelEncoder()
for c in cat:
    df_suicide_tree[c] = l.fit_transform(df_suicide_tree[c])
```

[604]:
```
df_suicide_tree
```

[604]:
```
       country  year  sex  age  suicides_no  population  suicides/100k pop  \
0            0     2    1    0           21      312900               6.71
1            0     2    1    2           16      308000               5.19
2            0     2    0    0           14      289700               4.83
3            0     2    1    5            1       21800               4.59
4            0     2    1    1            9      274300               3.28
…          …   …  …  …           …          …                  …
27815      100    29    0    2          107     3620833               2.96
27816      100    29    0    5            9      348465               2.58
27817      100    29    1    3           60     2762158               2.17
27818      100    29    0    3           44     2631600               1.67
27819      100    29    0    4           21     1438935               1.46

       gdp_for_year ($)   gdp_per_capita ($)  generation
0          2,156,624,900                 796           2
1          2,156,624,900                 796           5
2          2,156,624,900                 796           2
3          2,156,624,900                 796           1
```

```
4          2,156,624,900                  796           0
…                    …                    …           …
27815     63,067,077,179                 2309           2
27816     63,067,077,179                 2309           5
27817     63,067,077,179                 2309           3
27818     63,067,077,179                 2309           3
27819     63,067,077,179                 2309           0

[27820 rows x 10 columns]
```

[605]:
```
df_suicide_tree.columns
```

[605]:
```
Index(['country', 'year', 'sex', 'age', 'suicides_no', 'population',
       'suicides/100k pop', ' gdp_for_year ($) ', 'gdp_per_capita ($)',
       'generation'],
      dtype='object')
```

[606]:
```
df_suicide_tree.dtypes
```

[606]:
```
country                 int32
year                    int64
sex                     int32
age                     int32
suicides_no             int64
population              int64
suicides/100k pop     float64
 gdp_for_year ($)      object
gdp_per_capita ($)      int64
generation              int32
dtype: object
```

[607]:
```
df_suicide_tree[' gdp_for_year ($) '] = df_suicide_tree[' gdp_for_year ($) '].
 ↪str.replace(',','').astype(float)
```

[608]:
```
n = ['suicides_no', 'population', 'suicides/100k pop', ' gdp_for_year ($)␣
 ↪','gdp_per_capita ($)']
r = RobustScaler()
df_suicide_tree[n] = r.fit_transform(df_suicide_tree[n])
```

[609]:
```
df_suicide_tree
```

[609]:
```
        country  year  sex  age  suicides_no  population  suicides/100k pop  \
0             0     2    1    0    -0.031250   -0.084435           0.045860
1             0     2    1    2    -0.070312   -0.087963          -0.050955
2             0     2    0    0    -0.085938   -0.101142          -0.073885
3             0     2    1    5    -0.187500   -0.294064          -0.089172
4             0     2    1    1    -0.125000   -0.112232          -0.172611
```

```
…         …   …  …  …          …          …                 …
27815      100    29    0    2     0.640625    2.297696          -0.192994
27816      100    29    0    5    -0.125000   -0.058824          -0.217197
27817      100    29    1    3     0.273438    1.679341          -0.243312
27818      100    29    0    3     0.148438    1.585323          -0.275159
27819      100    29    0    4    -0.031250    0.726453          -0.288535

        gdp_for_year ($)   gdp_per_capita ($)  generation
0               -0.182942           -0.400243           2
1               -0.182942           -0.400243           5
2               -0.182942           -0.400243           2
3               -0.182942           -0.400243           1
4               -0.182942           -0.400243           0
…                   …                 …          …
27815            0.059520           -0.329631           2
27816            0.059520           -0.329631           5
27817            0.059520           -0.329631           3
27818            0.059520           -0.329631           3
27819            0.059520           -0.329631           0

[27820 rows x 10 columns]
```

Creating 4 classes based on suicide rate: Very low, Low, Medium, and High

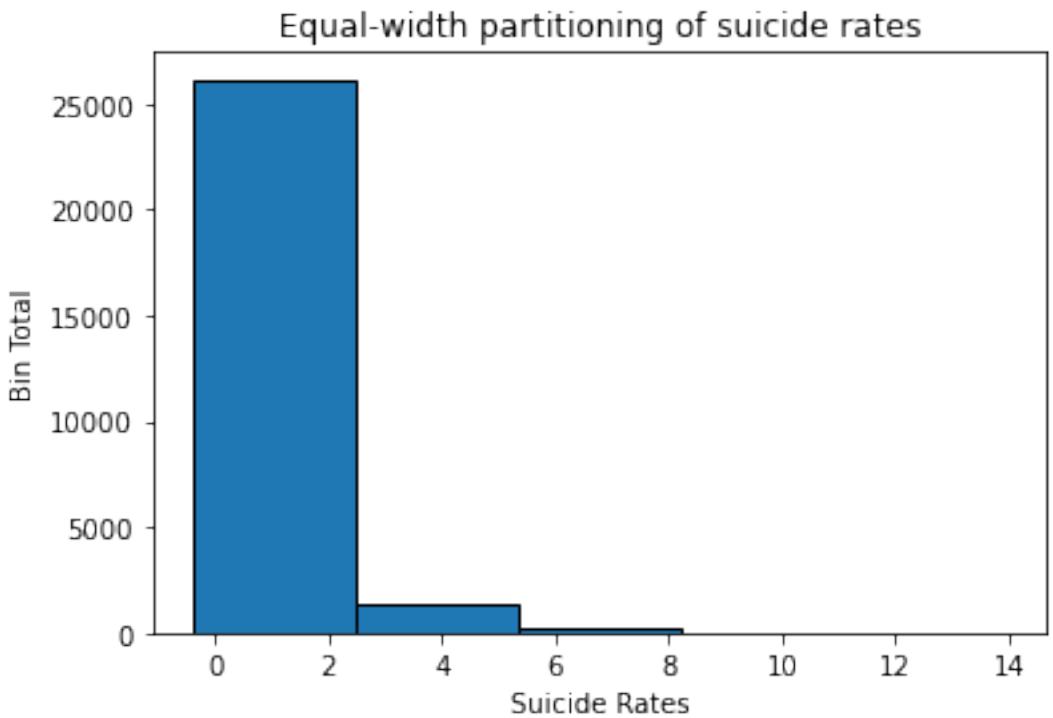
[610]:
```
df_suicide_tree.describe()
```

[610]:
```
            country          year           sex           age   suicides_no  \
count  27820.000000  27820.000000  27820.000000  27820.000000  27820.000000
mean      49.275270     16.258375      0.500000      2.499425      1.699800
std       29.372538      8.469055      0.500009      1.708754      7.047249
min        0.000000      0.000000      0.000000      0.000000     -0.195312
25%       24.000000     10.000000      0.000000      1.000000     -0.171875
50%       47.000000     17.000000      0.500000      2.000000      0.000000
75%       74.000000     23.000000      1.000000      4.000000      0.828125
max      100.000000     31.000000      1.000000      5.000000    174.320312

         population  suicides/100k pop   gdp_for_year ($)   \
count  27820.000000       27820.000000        27820.000000
mean       1.018722           0.434783            1.582163
std        2.816976           1.207740            5.786271
min       -0.309562          -0.381529           -0.191340
25%       -0.239551          -0.322930           -0.155759
50%        0.000000           0.000000            0.000000
75%        0.760449           0.677070            0.844241
max       31.235537          13.947771           71.940171

       gdp_per_capita ($)    generation
count        27820.000000  27820.000000
```

```
mean            0.349767      2.701869
std             0.881485      1.809836
min            -0.425678      0.000000
25%            -0.276520      1.000000
50%             0.000000      2.000000
75%             0.723480      4.000000
max             5.459467      5.000000
```

[611]:
```
suicides_list = df_suicide_tree['suicides/100k pop'].tolist()
suicides_list = sorted(suicides_list)
```

[612]:
```
plt.hist(suicides_list, bins = 5, edgecolor = "black")
plt.title("Equal-width partitioning of suicide rates")
plt.xlabel("Suicide Rates")
plt.ylabel("Bin Total")
plt.show()
```



[613]:
```
suicides_list_range = round(max(suicides_list) - min(suicides_list), 2)
print("The range is", suicides_list_range)
length = round((suicides_list_range / 5), 2)
```

```
The range is 14.33
```

[614]:
```
cut_1 = round(min(suicides_list) + length, 2)
print("Cut 1 is", cut_1)
cut_2 = round(cut_1 + length, 2)
print("Cut 2 is", cut_2)
cut_3 = round(cut_2 + length, 2)
print("Cut 3 is", cut_3)
```

```
Cut 1 is 2.49
Cut 2 is 5.36
Cut 3 is 8.23
```

[615]:
```
Class1 = df_suicide_tree[df_suicide_tree['suicides/100k pop'] <= 2.49]
Class1['Class'] = "Very Low"
Class2 = df_suicide_tree[(df_suicide_tree['suicides/100k pop'] > 2.49) &␣
↪(df_suicide_tree['suicides/100k pop'] <= 5.36)]
Class2['Class'] = "Low"
Class3 = df_suicide_tree[(df_suicide_tree['suicides/100k pop'] > 5.36) &␣
↪(df_suicide_tree['suicides/100k pop'] <= 8.23)]
Class3['Class'] = 'Medium'
Class4 = df_suicide_tree[df_suicide_tree['suicides/100k pop'] > 8.23]
Class4['Class'] = 'High'
```

[616]:
```
Concat1 = pd.concat([Class1, Class2], axis=0)
Concat2 = pd.concat([Concat1, Class3], axis=0)
suicide_tree = pd.concat([Concat2, Class4], axis=0)
```

[617]:
```
suicide_tree
```

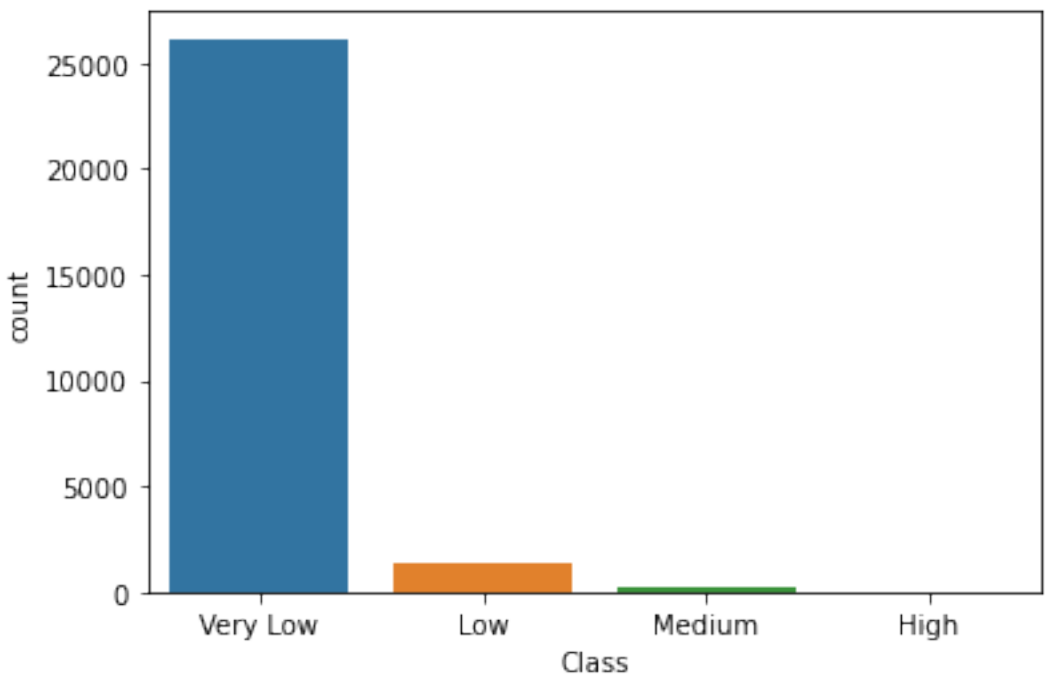
[617]:
```
       country  year  sex  age  suicides_no  population  suicides/100k pop  \
0            0     2    1    0    -0.031250   -0.084435           0.045860
1            0     2    1    2    -0.070312   -0.087963          -0.050955
2            0     2    0    0    -0.085938   -0.101142          -0.073885
3            0     2    1    5    -0.187500   -0.294064          -0.089172
4            0     2    1    1    -0.125000   -0.112232          -0.172611
…          …    …  …    …           …          …                  …
22400       82     0    1    5    -0.015625   -0.298312           8.831847
22436       82     3    1    5     0.039062   -0.294208           8.464968
23108       84    16    1    5     0.156250   -0.287369           8.835669
24092       88     5    1    5    -0.171875   -0.308322           9.172611
24332       88    27    1    5    -0.117188   -0.305913          11.533121

        gdp_for_year ($)   gdp_per_capita ($)  generation     Class
0               -0.182942           -0.400243           2  Very Low
1               -0.182942           -0.400243           5  Very Low
2               -0.182942           -0.400243           2  Very Low
3               -0.182942           -0.400243           1  Very Low
4               -0.182942           -0.400243           0  Very Low
…                     …                   …          …        …
```

```
22400          -0.115344          -0.057404          1      High
22436          -0.090668           0.056657          1      High
23108          -0.108429           0.074999          5      High
24092          -0.189981          -0.385402          1      High
24332          -0.171703           0.039903          5      High

[27820 rows x 11 columns]
```

Checking and fixing imbalanced classes

[618]:
```
ax = sns.countplot(x=suicide_tree['Class'], data=suicide_tree)
print(suicide_tree['Class'].value_counts())
```

```
Very Low    26159
Low          1362
Medium        265
High           34
Name: Class, dtype: int64
```



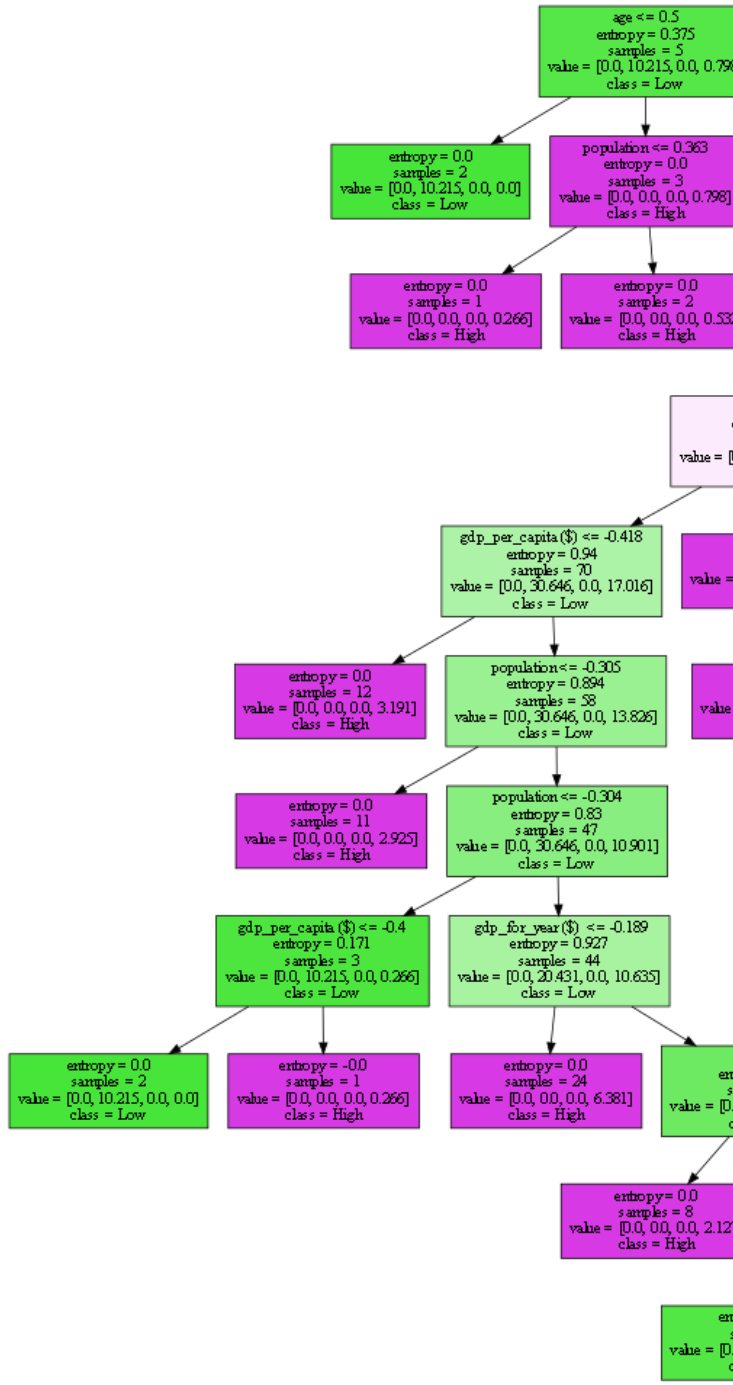
DecisionTree Classifier 2

[644]:
```
y = suicide_tree['Class']
x = suicide_tree.drop(['Class', 'suicides/100k pop', 'suicides_no', 'country'],␣
 ↪axis=1)

X_train, X_test, Y_train, Y_test = train_test_split(x, y, test_size=0.4,
```

```
                                                        random_state=1,
                                                       ␣
 ↪stratify=suicide_tree['Class'])
clf_train = tree.DecisionTreeClassifier(criterion='entropy',␣
 ↪class_weight='balanced', max_depth=14)
clf_train = clf_train.fit(X_train, Y_train)

dot_data_train = tree.export_graphviz(clf_train, feature_names=x.columns,
                                      class_names=['Very Low', 'Low','Medium',␣
 ↪'High'],
                                      filled=True, out_file=None)
graph_train = pydotplus.graph_from_dot_data(dot_data_train)
Image(graph_train.create_png())
```

```
dot: graph is too large for cairo-renderer bitmaps. Scaling by 0.564102 to fit
```

[644]:


age <= 0.5
entropy = 0.375
samples = 5
class = Low
entropy = 0.0
samples = 2
class = Low
population <= 0.363
entropy = 0.0
samples = 3
class = High
entropy = 0.0
samples = 1
class = High
entropy = 0.0
samples = 2
class = High
gdp_per_capita ($) <= -0.418
entropy = 0.94
samples = 70
class = Low
entropy = 0.0
samples = 12
class = High
population <= -0.305
entropy = 0.894
samples = 58
class = Low
entropy = 0.0
samples = 11
class = High
population <= -0.304
entropy = 0.83
samples = 47
class = Low
gdp_per_capita ($) <= -0.4
entropy = 0.171
samples = 3
class = Low
gdp_for_year ($) <= -0.189
entropy = 0.927
samples = 44
class = Low
entropy = 0.0
samples = 2
class = Low
entropy = -0.0
samples = 1
class = High
entropy = 0.0
samples = 24
class = High
entropy = 0.0
samples = 8
class = High

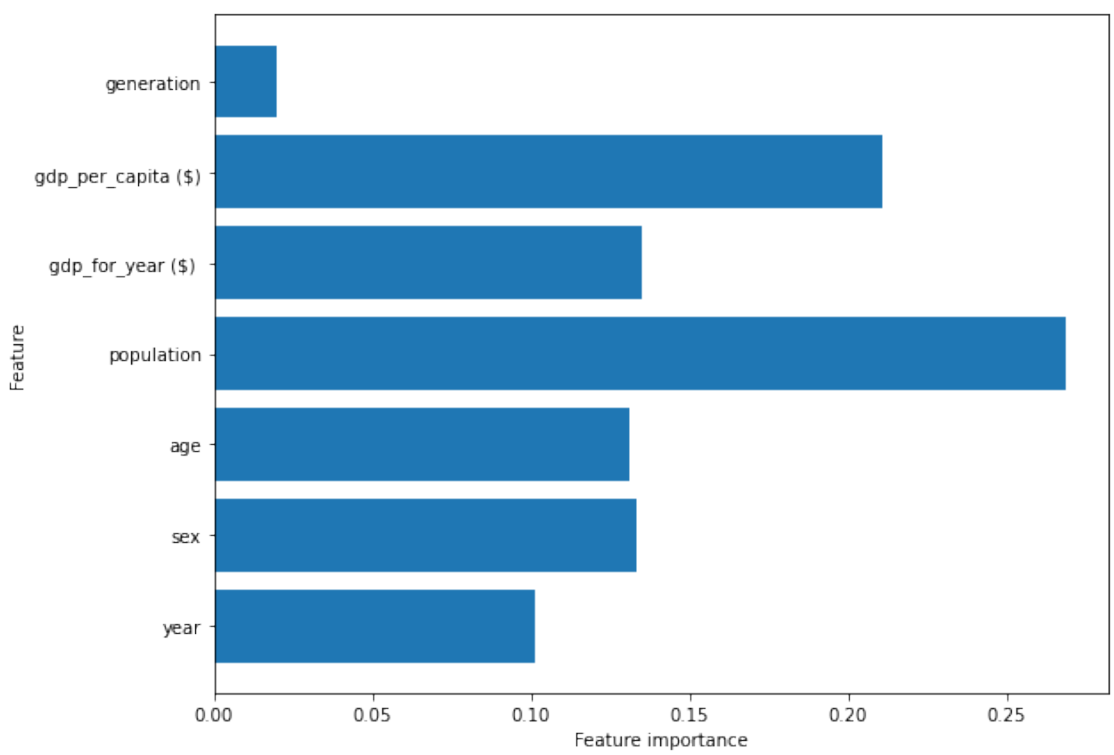
Checking accuracy and the features importance

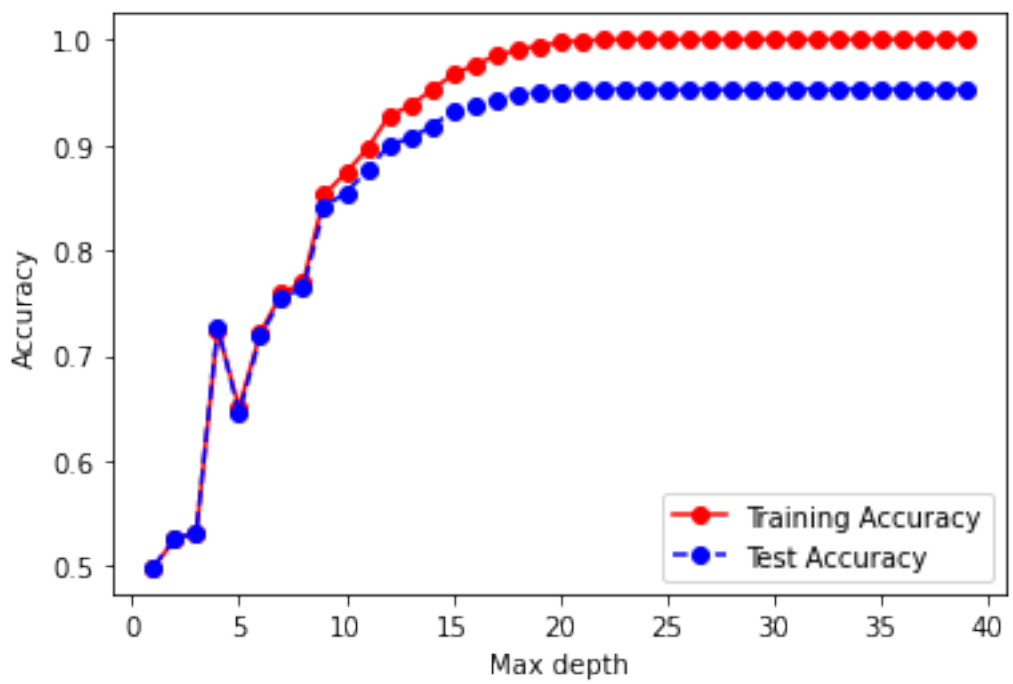
[645]:
```
Y_predTrain = clf_train.predict(X_train)
Y_predTest = clf_train.predict(X_test)
trainAcc = accuracy_score(Y_train, Y_predTrain)
testAcc = accuracy_score(Y_test, Y_predTest)
print('Accuracy on train data is %.2f' % trainAcc)
print('Accuracy on test data is %.2f' % testAcc)
```

```
Accuracy on train data is 0.95
Accuracy on test data is 0.91
```

Note that the accuracy will increase if we increase the training size (this accuracy 0.91 based on 0.6 training , 0.4 test_size)

[621]:
```
plt.figure(figsize=(9,7))
n_features = X_train.shape[1]
plt.barh(range(n_features), clf_train.feature_importances_, align='center')
plt.yticks(np.arange(n_features), X_train.columns)
plt.xlabel("Feature importance")
plt.ylabel("Feature")
plt.show()
```



[622]:
```
L=[i for i in np.arange(1,160,5)]
maxdepths = [i for i in np.arange(1,40,1)]
trainAcc = np.zeros(len(maxdepths))
testAcc = np.zeros(len(maxdepths))
index = 0
for depth in maxdepths:
    clf = tree.DecisionTreeClassifier(class_weight='balanced', max_depth=depth)
    clf = clf.fit(X_train, Y_train)
    trainAcc[index] = accuracy_score(Y_train, clf.predict(X_train))
    testAcc[index] = accuracy_score(Y_test, clf.predict(X_test))
    index += 1
plt.plot(maxdepths,trainAcc,'ro-',maxdepths,testAcc,'bo--')
plt.legend(['Training Accuracy','Test Accuracy'])
plt.xlabel('Max depth')
plt.ylabel('Accuracy')
```

[622]: Text(0, 0.5, 'Accuracy')